Integration Guide
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Integration Guide for the Offer-to-Cash End-to-End Process Based on SAP Hybris Billing

SAP S/4HANA 2017



Document History

Caution

Before you start the implementation, make sure you have the latest version of this document. You can find the latest version at the following location: service.sap.com/BRIM .

The following table provides an overview of the most important document changes.

Table 1

Version	Date	Description
1.0	2016-12-05	Initial version for SAP S/4HANA 1511
1.1	2017-01-25	Point “Migration from Direct to Joint Master Data Distribution” added

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1 Purpose of this Document

This document is intended as the central starting point for the Offer-to-Cash end-to-end process. It gives you an overview of the Offer-to-Cash end-to-end process and guides you through the activities that are necessary to connect the components that make up the Offer-to-Cash scenario landscape. The document lists all required activities for technical and process integration. For more detailed information, see the following sources:

- Configuration documentation in SAP Solution Manager
- SAP Library documentation on the SAP Help Portal
- Implementation Guides (IMG) in the system

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2 The Offer-to-Cash End-to-End Process

SAP Hybris Billing, formerly known as SAP Billing and Revenue Innovation Management (BRIM), is designed to streamline your high-volume revenue management processes and consists of the following components:

- SAP CRM containing SAP Hybris Billing, subscription order management, and parts of SAP Hybris Billing, customer financials
- SAP Hybris Billing, mediation by DigitalRoute
- SAP Hybris Billing, charging
- SAP Hybris Billing, pricing simulation
- SAP ERP Business Suite or SAP S/4 HANA containing SAP Hybris Billing, invoicing and parts of SAP Hybris Billing, customer financials

The following graphic illustrates which solution capability is provided by which solution component of SAP Hybris Billing.

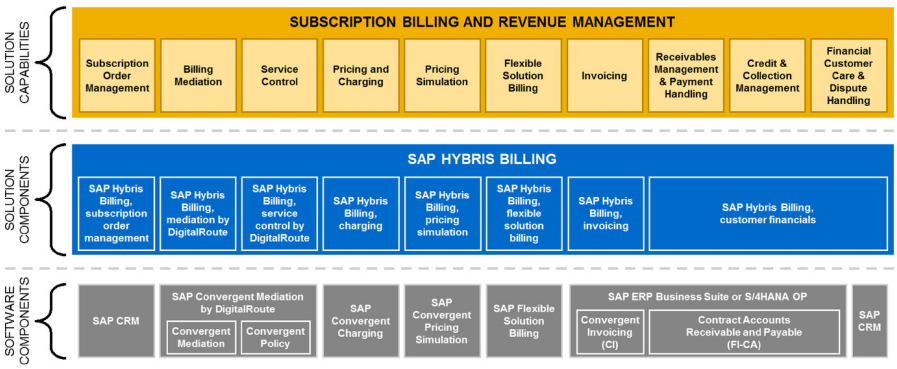


Figure 1: Solution Overview of the SAP Hybris, Billing Components

While SAP Convergent Charging (SAP Hybris Billing, charging) and SAP CRM each represent separate products, Contract Accounts Receivable and Payable (FI- CA), which includes SAP Convergent Invoicing (SAP Hybris Billing, invoicing), is part of the following industry solutions:

- SAP Contract Accounts Receivable and Payable
- SAP for Telecommunications
- SAP Utilities
- SAP Public Sector Collection and Disbursement (PSCD)

SAP Hybris Billing is modular, so you have the option of deploying single elements within your existing IT software landscape.

Rapid-Deployment Solutions

SAP provides the SAP Billing and Revenue Innovation Management rapid-deployment solution (BRIM), which joins the SAP Customer Relationship Management application, the SAP Hybris Billing, charging application, the

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SAP Hybris Billing, invoicing package, and the SAP Hybris Billing, customer financial package to create a comprehensive solution that supports sales order and contract creation and the consume-to-cash business process.

SAP Billing and Revenue Innovation Management rapid-deployment solution: service.sap.com/public/rds

For a specific planning, risk management, and performance optimization processes for individual industries SAP provides the SAP Hybris Marketing rapid-deployment solution (service.sap.com/public/rds-ymkt)

This application allows you to gain enhanced visibility on service usage of customers through multiple visualization layers and quickly deploy predefined views and dashboards as a basis for your specific analysis.

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3 Prerequisites of the Offer-to-Cash End-to-End Process

Offer-to-Cash supports the following release combinations of its components:

Table 2

	SAP S/4 HANA	SAP CRM	SAP Convergent Charging
SAP Business Suite 4 SAP HANA	SAP S/4HANA 1511	SAP enhancement package 4 for SAP CRM 7.0	SAP CC 4.1
SAP Business Suite 4 SAP HANA	SAP S/4HANA 1610	SAP enhancement package 4 for SAP CRM 7.0	SAP CC 4.1

> **Note**
>
> This guide deals only with the functionality provided by SAP S/4HANA 1511 and SAP S/4HANA 1610
>
> This guide does not cover the Fulfillment process, that is, Sales and Distribution, Logistics, 3rd party service provisioning and activation, and workforce management.

Depending on the industry solution you are using, the following business function sets and business functions must be activated in the S/4 HANA system:

Table 3

SAP Telecommunications	Business function set SAP Telecommunications (TELCO) with all its business functions
SAP Contract Accounts Receivable and Payable (FI-CA)	Business function set SAP Contract Accounts Receivable and Payable (FICAX) with all its business functions
SAP Utilities	The following business functions of business function set Industry Solution for Utilities (`UTILITIES`): • `ISU_UTIL_1` • `ISU_UTIL_WASTE` • `ISU_LOC_CI_1` • `ISU_CA_1` • `ISU_CA_2` • `ISU_CA_3` • `ISU_CA_3B` • `ISU_CA_3E` • `ISU_CA_3G` • `ISU_INV_1` • `ISU_INV_2` • `ISU_INV_2B`

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	• ISU_INV_2F • ISU_INV_2G • ISU_INV_2H • ISU_INV_2J • ISU_INV_2K • ISU_INV_PP_1 • ISU_INV_PP_2 • ISU_INV_PP_3 • ISU_INV_PP_3A • ISU_INV_PP_3D • ISU_INV_PP_3E
SAP Public Sector Collection and Disbursement	• PSCD_COLLECTIONS_MANAGEMENT • PSCD_COLLECTIONS_MANAGEMENT_1 • PSCD_COLLECTIONS_MANAGEMENT_2 • PSCD_COLLECTIONS_MANAGEMENT_3 • PSCD_COLLECTIONS_MANAGEMENT_3C • PSCD_COLLECTIONS_MANAGEMENT_3D • PSCD_COLLECTIONS_MANAGEMENT_4 • PSCD_COLLECTIONS_MANAGEMENT_4B • PSCD_COLLECTIONS_MANAGEMENT_4E • PSCD_COLLECTIONS_MANAGEMENT_4F • PSCD_COLLECTIONS_MANAGEMENT_4G • PSCD_COLLECTIONS_MANAGEMENT_5 • PSCD_INV_PP_1 • PSCD_INV_PP_1D (• PSCD_INV_PP_1E

If you want to use the integration the Offer-to-Cash business process provides between SAP S/4 HANA and SAP CRM, the following **business functions** must be activated in the **CRM system**:

- CRM_SLS_SRV_1
- CRM_TEL_SOM_1
- CRM_PROVORDERINT_1
- CRM_PROVORDERINT_2
- CRM_PROVORDERINT_3_2
- CRM_PROVORDERINT_3_4
- CRM_PROVORDERINT_3_5
- CRM_PROVORDERINT_3_7

  CRM_PROVORDERINT_3_8

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- CRM_PROVORDERINT_3_9
- CRM_PROVORDERINT_4
- CRM_PROVORDERINT_4_2

If you are using SAP Telecommunications or SAP Contract Accounts Receivable and Payable (FI-CA) and want to use Financial Customer Care und Dispute Management within the Offer-to-Cash business process, the business functions CRM_FCA_1 and CRM_FCA_2 must be activated in the CRM system.

If you are using SAP Utilities and want to use Financial Customer Care und Dispute Management within the Offer-to-Cash business process, the business function CRM_UT_ER_2 must be activated in the CRM system.

For more information on how to use Financial Customer Care und Dispute Management, also see SAP Note 746475.

> **Note**
>
> The business functions provided with SAP CRM are not industry-dependent but are available to all industries offering the Offer-to-Cash end-to-end process.
>
> The business functions and business function sets listed for each industry solution in the table above are a prerequisite for the mentioned CRM business functions.

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4 Products

4.1 Modelling Products

Product modeling is logically divided into the following views:

- Commercial view of products

  SAP CRM contains the master of the commercial view of all products. This master comprises, for example, the definition of options available for customer selection, bundling of different services, and the definition of one-off charges (for example, activation fees, early termination fees, change fees).

- Charging view of products

  SAP Convergent Charging charges for the consumption of services by end customers. The SAP Convergent Charging system acquires and collects consumption information in real time (chargeable items) or in batch mode (consumption data records). In order to charge for the consumption of a service, you have to define charge plans in SAP Convergent Charging.

  In order to process refills in prepaid scenarios, you have to define refill plans in SAP Convergent Charging.

  Charge plans and refill plans are used at run-time to perform the rating, charging and refilling processes for consumption and refill data. They contain the logic that dynamically determines the amounts and the account to be credited or debited for the consumption of a service or a refill, as well as the logic for generating temporarily charged items and billable items in Convergent Invoicing.

  During product definition in SAP CRM you assign charge plans and a single refill plan to a commercial product. They are referenced in each provider contract through the ordered product.

  For more information, see the section “Transactional Data Within the Offer-to-Cash Business Process” and the Configuration Guide for SAP Convergent Charging at help.sap.com/cc41 › *Configuration and Deployment Information* › *Configuration Guide (SAP Service Marketplace)* › *Install* › *SAP CC 4.1 Configuration Guide* › *Configuration with Integrated Scenario Consume-To-Cash* › *Master Data Basic Settings* › *Setting up the Master Data of the Service Provider*.

- Technical view of products

  To be able to activate the services your customers subscribe to and to process their usage, you have to assign the objects stored in the systems, which technically provide the services to the commercial view of products. SAP CRM provides a generic storage to manage the assignment of such services allowing you to store the IDs, descriptions and parameters describing these objects. First you define external object types. The external object type defines which kind of objects are available and which system they reside in. Based on this information you then assign external objects in the Cross Catalog Mapping (CCM). You can influence the activation of the external objects during contract replication by adding conditions to these assignments.

  For more information, see help.sap.com/crm704 › *Application Help* › *English* › *SAP CRM for Industries* › *Telecommunications* › *Functions for the Telecommunications* › *Functions for Provider Sales and Order Management* › *Products* › *Cross Catalog Management* › *CCM Assignment of External Objects*.

  For the integration at contract level, create a custom step in the Order Distribution Infrastructure (ODI) in SAP CRM calling the API of the external system. SAP CRM provides the tool class `CL_CRM_ISX_CCM_EXTOBJ_MAP` you can use to determine the appropriate external objects and the correct values for the parameters at the contract level based on the definition set up in the Cross Catalog Mapping (CCM).

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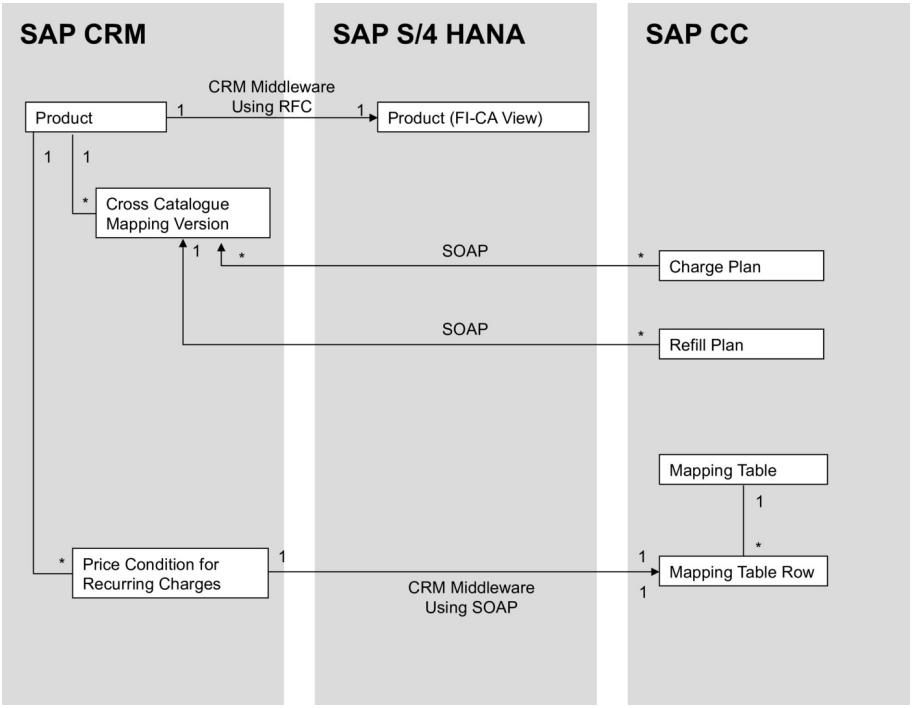


Figure 2: Products Within the Offer-to-Cash Business Process

The figure above shows the different steps of product definition:

- Commercial product definition in SAP CRM with replication to Contract Accounts Receivable and Payable (FI-CA)
- Cross-catalog mapping to SAP Convergent Charging and the definition of prices

The commercial product created in the CRM system is replicated to the S/4 HANA system and contains only the data required in the S/4 HANA system for data processing. In the S/4 HANA system, the product is used to centrally store product texts as well as account determination attributes.

In order to distribute products between the S/4 HANA and CRM system, configure the CRM system accordingly. For more information, see Customizing for Customer Relationship Management under *Cross Industry Functions* > *Master Data* > *Products* > *Settings for Product Replication to FI-CA*.

> **Note**
>
> The number ranges for the product types Material and Service are not allowed to overlap in the CRM system.
>
> Do not create any products with the same name for different product types.

The cross-catalog mapping in SAP CRM maps the charging view to the commercial view. The cross-catalog mapping defines which charge plans and refill plans must be active when customers sign contracts for a commercial product. In SAP CRM you also define the parameterization of the charge plans. The charge plans are not replicated from SAP Convergent Charging to the CRM system. Instead, the cross-catalog mapping stores references to the charge plans and to the underlying charge plan signature. The CRM system queries charge plans and imports them synchronously during the maintenance of the cross-catalog mapping. The same system behavior applies to refill plans.

The cross-catalog mapping also defines which allowances must be created when customers sign contracts for a commercial product or when changes are made to a customers' contract. SAP CRM creates allowances in SAP CC either on demand or automatically when the provider contract is activated or changed. When SAP CRM creates an allowance, it sends a charging request to SAP CC using the chargable item class `ALLO` (see **1** in the

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figure below). The chargeable item contains the identifier of the allowance definition to be created and the date at which the allowance must be activated. The charging request executes a SAP CC charge (see **2** in the figure below). This charge retrieves the corresponding allowance definition from the allowance definition table, creates the corresponding allowance, and initializes the allowance according to the allowance definition (see **3** in the figure below). The charge can generate a billable item for reporting purposes (see 4 in the figure below). The allowance can generate billable items that trigger the posting of deferred revenue (see **4** in the figure below).

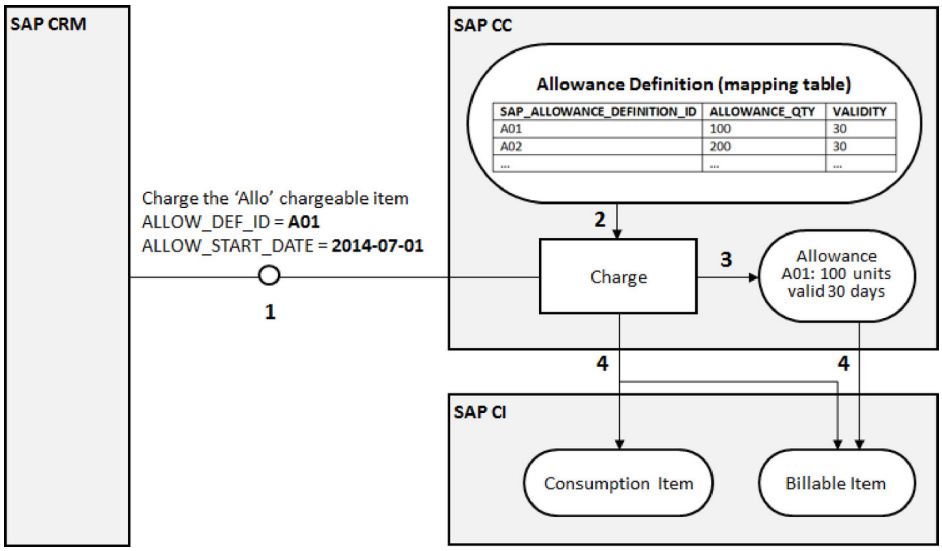


Figure 3: Integration between SAP CRM and SAP CC During Allowance Creation

To customize SAP CC, use the templates of the `ALLO` chargeable item class, the allowance definition table class, and the instance. For more information, see the section Allowances Within the Offer-to-Cash Business Process [page 40].

Technical Configuration

In order for the CRM system to be able to read charge plans and refill plans from SAP Convergent Charging, you have to configure the consumer proxy `CO_CRM_ISX_CC_CATALOG_SERVICES` as described in the section "Setting Up Communication with SAP Convergent Charging Using Web Services".

For more information, see the section "Cross Catalog Mapping" of the configuration guide in SAP Solution Manager under *Solutions/Applications* > *Cross Industry Scenarios* > *Scenarios* > *Offer to Fulfill* > *Sales and Order Management* on the *Configuration* tab.

For each product you define in SAP CRM, the system creates a mapping table in SAP Convergent Charging. For each price condition for recurring charges within the product, the system creates one row within this mapping table.

You maintain prices according to the following rules:

- You maintain usage prices in SAP Convergent Charging.
- You maintain one-off prices in SAP CRM.
- You maintain recurring prices in SAP CRM as pricing condition records.

  They are replicated to SAP Convergent Charging to mapping tables and mapping table records, as SAP Convergent Charging applies the recurring charges during each bill cycle. While SAP CRM only specifies prices for complete billing periods, SAP Convergent Charging can prorate prices if services have only been consumed during part of a bill cycle. As soon as the first price condition record is sent from CRM, a new mapping table is created for each CRM product in SAP Convergent Charging.

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In order to distribute prices from SAP CRM to SAP Convergent Charging, configure the consumer proxy `CO_CRM_ISX_CC_CATALOG_SERVICES` as described in the section “Setting Up Communication with SAP Convergent Charging Using Web Services”.

For more information, see the section “Integration of Recurring Charges with SAP” in the configuration guide in SAP Solution Manager under *Solutions/Applications* > *Cross Industry Scenarios* > *Scenarios* > *Offer to Fulfill* > *Sales and Order Management* on the *Configuration* tab.

> **Note**
>
> When you configure a product in the CRM system, you assign charge plans and refill plans to the product. The charge plans and refill plans you create in SAP Convergent Charging are not replicated in the CRM system. During product configuration, the CRM system queries the valid charge plans and refill plans from the SAP Convergent Charging system. The query returns only those charge plans and refill plans with the status released (which can no longer be modified except for a part of the mapping).

4.2 Transporting Products between Systems

To transport complex product structures between systems, as for example from your development system to your quality assurance system, please adhere to the following transport sequence:

1. Transport Products from Sales and Distribution (SD) in SAP S/4 HANA
2. Transport charge plans and tables from SAP CC
3. Transport business rules from SAP CRM
4. Transport products (services and materials) from SAP CRM

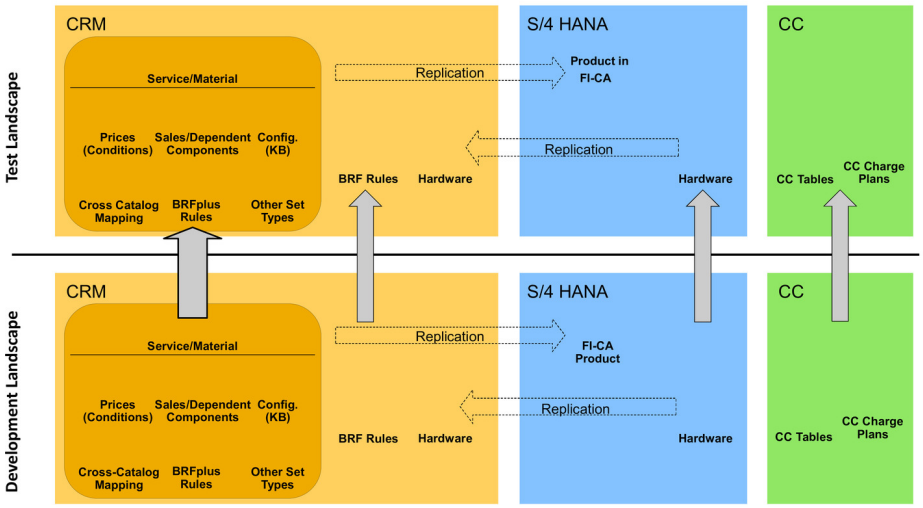


Figure 4: Overview of Product Transport

In SAP CRM you trigger the product transport from within the product maintenance. For more information, see the SAP CRM Application Help at help.sap.com/crm704 under *SAP CRM for Industries* > *Telecommunications* > *Functions for Provider Sales and Order Management* > *Product Transport*.

Remote function calls provide the communication between the different SAP systems. Make sure that the user you use in the source system is assigned the roles including all required authorizations for the Utilities product

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download. In the destination system, the user requires the same authorizations and, in addition, the authorization to create products.

To be able to transport product data between CRM systems, users require the appropriate authorization. To assign the required authorization, create an authorization definition for your SAP CRM users based on the `PFCG` role `SAP_CRM_PROVIDER_PROD_TRANSPRT` delivered by SAP. This role supplements the `SAP_CRM_UIU_PROVIDER_SALES` role.

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5 Business-Partner-Related Master Data

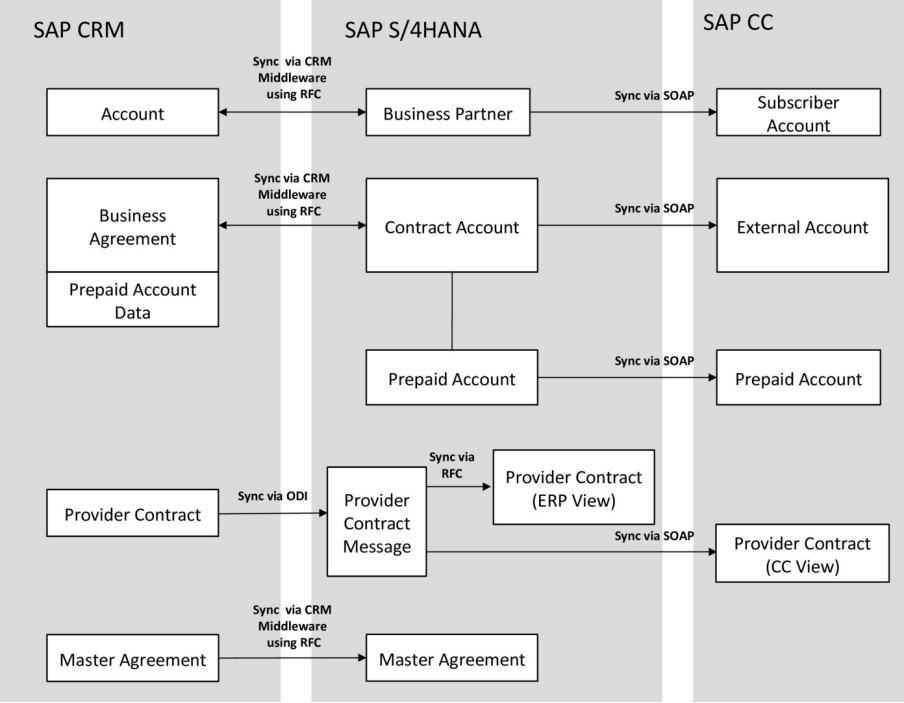


Figure 5: Business-Partner-Related Master Data in Offer-to-Cash

The figure above shows which master data objects are used by which components within Offer-to-Cash and how they interact:

- **SAP Customer Relationship Management**

  In the CRM system, you enter and manage customer and contract data. You transfer the relevant parts of this data that are needed in Contract Accounts Receivable and Payable and Convergent Invoicing to the SAP S/4HANA system.

  You maintain the prepaid account data on the level of the business agreement in SAP CRM. There can be only one prepaid account per business agreement.

  The account and business agreement as well as the master agreement in CRM are replicated to the SAP S/4HANA system using the CRM Middleware. The CRM Middleware makes sure that as soon as prepaid-relevant data is maintained in the business agreement, the data is transmitted to the prepaid account in the SAP S/4HANA system.

  The CRM system replicates the provider contract in the SAP S/4HANA system using the order distribution infrastructure (ODI). ODI calls the RFC `CRM_ISX_CONTRACT_MAINTAIN` to create the provider contract in the SAP S/4HANA as well as in the SAP Convergent Charging system. To supply the SAP Convergent Charging system with the relevant data, the RFC `CRM_ISX_CONTRACT_MAINTAIN` uses the Web service `ContractProvisioningServices`.

  In the CRM system use the distribution step types *Call ERP Message contract activation MCC* (`PCMA`) and *Call ERP MessageContractDeactivation MCC* (`PCMD`) in the corresponding distribution schema.

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For more information, see help.sap.com/crm704 > *Application Help* > *English* > *SAP CRM for Industries* > *Telecommunications* > *Functions for Provider Sales and Order Management* > *Provider Order and Provider Contract* > *Document Distribution* > *Joint Distribution of Master Data and Provider Contracts*.

Technical Configuration

You have to configure a logical port for the consumer proxy `CO_ISX_CC_CTR_PROV_SERVICES` as described in the section "Setting Up Communication with SAP Convergent Charging Using Web Services".

To ensure successful replication of provider contracts as well as master agreements, make sure that the number ranges used by SAP S/4HANA are synchronized with the number ranges used by SAP CRM.

Also make sure the Customizing for accounts and business partners, business agreements, and contract accounts, as well as prepaid accounts is the same in the SAP S/4HANA system and the CRM system.

For more information on how to replicate business partners in SAP for Telecommunications or SAP for Utilities, see service.sap.com/utilities > *SAP for Utilities - Product Information* > *SAP CRM for Utilities* > *Cookbooks & Guidelines* > *IS-U-specific Set-Up and Load Guide for Business Partner*.

For more information on how to replicate business partners in SAP Contract Accounts Receivables and Payables, see the application help at help.sap.com/fi-cax > *<Release>* > *Basic Functions* > *SAP Business Partner* > *Replication and Synchronization of Business Partners*.

You replicate master agreements using the CRM Middleware. In order to replicate master agreements, you use the adapter object `BUS_TRANS_MSG`. In the Middleware Administration Console (transaction `SMOEAC`), you must assign the subscription *All Business Transactions (MESG)* to the SAP S/4HANA site.

- **Migration from Direct to Joint Master Data Distribution**

  If you want to migrate to S/4HANA you need to use the joint master data distribution as this is the only replication method which is supported. If you are currently using the direct master data distribution, you need to migrate your existing distribution data before you can activate the joint master data distribution. The migration functionality is available from SAP ERP 6.0 EHP7 SPS14 and SAP ERP 6.0 EHP8 SPS06 onwards. Alternatively, if you are using older support packages, you can install the SAP Notes 2401583 and 2387353.

  Once the migration functionality is available in your system, proceed as follows to migrate from the direct to the joint master data distribution:

  1. Suspend the master data distribution in SAP CRM before you start with the migration. (That means business partner, business agreement/contract account, prepaid account, provider contract and master agreement creation and changes are not distributed to SAP ERP.)
  2. Call up transaction `FKKCC_MIG2JOINT` in SAP ERP. It consists of three steps but you can also execute the steps independently from each other (in this case, consider the step sequence!):
     1. Clean-up of distribution records

        The same functionality is provided by report `RFKK_CC_MD_CLEAN_UP`. This step is optional but recommended. By reducing the number of existing distribution records, the migration report will be executed faster.

     2. Creation of rating area assignment and distribution object data

        This step is mandatory. It creates rating area assignment records for business partners, contract accounts, and prepaid accounts for rating area `SPACE` and creates the distribution information for provider contracts. Please note that rating areas with value `SPACE` are already technically used for the joint master data replication although the functionality of rating areas (multiple SAP Convergent Charging instances) might not be active yet.

     3. Adaptation of distribution records

        This step is mandatory. All existing distribution records of business partners, contract accounts, and prepaid accounts that are not marked as obsolete will be adapted to the joint master data

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distribution. In addition, if no rating area assignment exists for those objects up to this point, it will be created. In this step, errors can be thrown if the object to be processed is currently locked by another process. In this case, you need to execute this step again when the object lock is removed so that everything is properly migrated to the joint master data distribution.

Make sure that no errors are shown in the application log of the transaction. Execute the transaction again until all errors of the affected step are resolved.

3. Activate the joint master data distribution in SAP CRM and SAP ERP.
4. Resume the master data distribution of business partners, business agreements/contract accounts, prepaid accounts, provider contracts, and master agreements in SAP CRM.

> **Note**
>
> If erroneously distributed provider contracts still exist in SAP CRM from the direct master data distribution, you can resolve this by choosing “Cancel Activation ...and Return to Order Processing” in the ODI distribution monitor. Afterwards, you can submit the order once again. Then, the distribution will be done by using the joint master data distribution.

- **SAP S/4HANA**

  The relationship between the business partner and contract account depends on the setting of the contract account category.

  In Customizing for Contract Accounts Receivable and Payable under *Basic Functions* > *Contract Accounts* > *Number Ranges and Contract Account Categories* > *Configure Contract Acct Categories and Assign Number Ranges*, you define for each contract account category whether only one or whether more than one business partner can be assigned to a contract account.

  The prepaid account is a separate object which is linked with the business partner and contract account.

  The SAP S/4HANA system transfers the business partner, contract account, and prepaid account data to the SAP Convergent Charging system by means of the Web service `SUACProvisioningServices` using the simple object access protocol (SOAP) and creates the corresponding subscriber account, external account, and prepaid account in SAP CC.

  The complete provider contract message is stored in Contract Accounts Receivable and Payable (FI-CA) in the SAP S/4HANA system. When the contract is created in Contract Accounts Receivable and Payable or SAP Convergent Charging each component only stores those attributes relevant for further processing within that component.

  As a prerequisite, you have to activate replication in Customizing for Contract Accounts Receivable and Payable under *Integration* > *Convergent Charging* > *Activate Joint Replication of All Master Data*.

  If you are using SAP CRM, enter the function module `FKK_SAMPLE_5707_SAPCRM` for event 5707 in Customizing for Contract Accounts Receivable and Payable under *Program Enhancements* > *Define Customer-Specific Function Modules*. The function module informs SAP CRM about the provider contract status.

  If you connect multiple FI-CA systems to one SAP CC system and you activate this option, the SAP S/4HANA system specifies the customer management area (`SAP_CMA`) in the subscriber account using the function module `LCR_GET_OWN_BUSINESS_SYSTEM`. The technical name of the customer management area consists of a combination of the `SID` and the client number.

  Technical Configuration

  You have to configure a logical port for the consumer proxy `CO_FKKCC_SUBSCRIBER_ACCOUNT` as described in the section “Setting Up Communication with SAP Convergent Charging Using Web Services”.

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Note

If you want to implement the scenario without using SAP CRM, you can maintain business partners, contract accounts, prepaid accounts, master agreements, and products in the SAP S/4HANA system using methods of the following BOR object types (transaction `SWO1`):

Table 4

Component	Object	BOR Object Type
Contract Accounts Receivable and Payable (FI-CA)	Business Partner	`CA_BUS1006`
SAP for Telecommunications (IS-T) SAP for Utilities (IS-U)	Business Partner	`ISUPARTNER`
Contract Accounts Receivable and Payable (FI-CA)	Contract Account	`CA_CONTACC`
SAP for Telecommunications (IS-T) SAP for Utilities (IS-U)	Contract Account	`ISUACCOUNT`
Contract Accounts Receivable and Payable (FI-CA)	Prepaid Account	`CA_PREPACC`
Contract Accounts Receivable and Payable (FI-CA)	Master Agreement	`CA_MSTRAGR`
Contract Accounts Receivable and Payable (FI-CA)	Product	`CA_BILLPRD`

You can maintain provider contracts using the RFC `CRM_ISX_CONTRACT_MAINTAIN`.

For more information on the interfaces existing between SAP Convergent Charging, SAP CRM and SAP S/4HANA, see Customizing for Contract Accounts Receivable and Payable under *Integration* > *Convergent Charging* > *Manage Interfaces to SAP CC* > *Display Overview of Interfaces to SAP CC*.

In the interests of efficient data storage and distribution, all your customer master data is jointly replicated to SAP CC. This means that the business partner, contract accounts, and prepaid accounts are not distributed immediately from the S/4 HANA to the SAP CC system. Instead, they are not distributed until the data of the provider contract they belong to is received in the S/4 HANA system.

The S/4 HANA system persists the master data for the provider contract and distributes it to the SAP CC system. In this way, you can monitor the distribution of all master data, as well as handle errors in the S/4 HANA system.

The S/4 HANA system ensures the logical order of replication of all master data objects from the S/4 HANA to the SAP CC system.

Note

Before you can connect multiple SAP CC systems to an S/4 HANA system you have to activate the joint master data distribution in Customizing for Contract Accounts Receivable and Payable.

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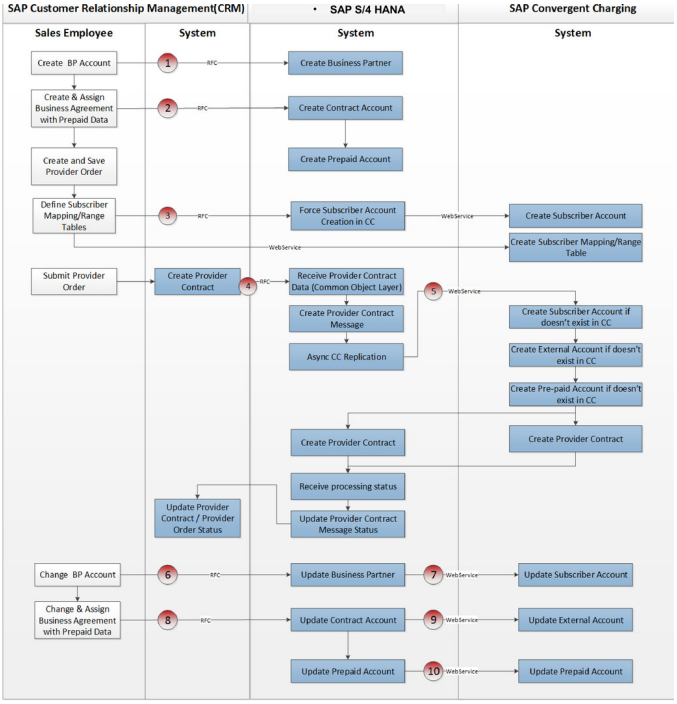


Figure 6: Joint Distribution of New Business Partners and Provider Contracts and Business Partner and Business Agreement Changes Through the System Landscape

Master data for new business partners and provider contracts without subscriber mapping tables or range tables is replicated in the following order:

1. Replication of business partner from the CRM to the S/4 HANA system
2. Replication of business agreements (with prepaid data) from the CRM to the S/4 HANA system
3. Replication of provider contract messages from the CRM to the S/4 HANA system

   The provider contract message contains all S/4 HANA and SAP CC data of a provider contract.

> **Note**
>
> At this stage, the provider contract has not yet been created in the S/4 HANA or SAP CC system.

4. Replication of business partners from the S/4 HANA to the SAP CC system
5. Replication of contract accounts from the S/4 HANA to the SAP CC system
6. Replication of prepaid accounts from the S/4 HANA to the SAP CC system
7. Replication of provider contracts from the CRM to the S/4 HANA and SAP CC systems
8. S/4 HANA system sends the status of the replication to the CRM system

If you are maintaining subscriber mapping tables or range tables in the CRM system, the business partner is replicated to the SAP CC system immediately. After that, the subscriber mapping table or range table is replicated to the SAP CC system. Changes for a business partner, contract account or prepaid account are immediately replicated from the S/4 HANA to the SAP CC system.

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5.1 Master Data Replication to Multiple SAP CC Systems

You can connect multiple SAP CC systems to an S/4 HANA system.

To be able to connect multiple SAP CC systems to an S/4 HANA system:

- You have set up joint master data distribution in the S/4 HANA system as described in chapter 5.1.
- You have activated multiple SAP CC systems for your S/4 HANA system in Customizing for Contract Accounts Receivable and Payable under *Integration* > *Convergent Charging* > *Specifications for Connecting Multiple CC Systems to an FI-CA System*.

To be able to connect multiple SAP CC systems to a CRM system:

- You have set up joint master data distribution in the CRM system as described in chapter 5.1.
- You have maintained the settings in the activity *Settings for Joint Distribution and Multiple SAP CC Systems* in Customizing for Customer Relationship Management under *Cross-Industry Functions* > *Provider Order and Contract Management* > *Joint Distribution and Multiple SAP CC Systems for a SAP CRM System*.

Rating areas are used to connect multiple SAP CC systems to your S/4 HANA and CRM systems. The following figure illustrates how rating areas are used.

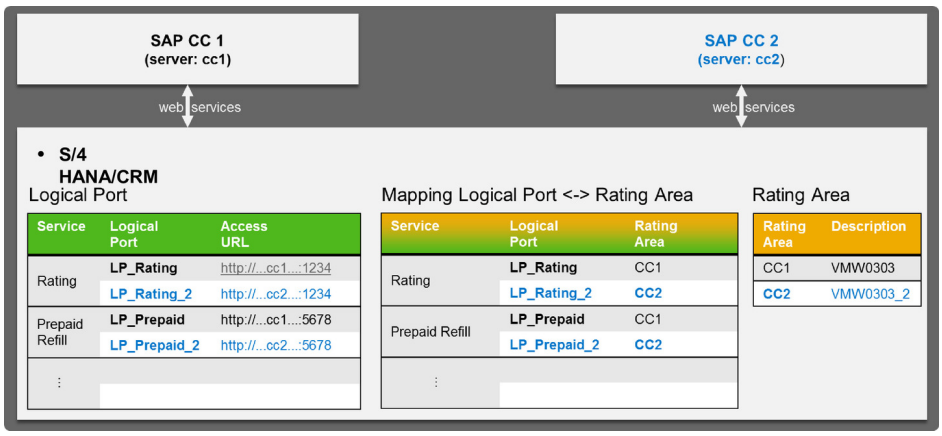


Figure 7: Configuring Rating Areas

Each SAP CC system is represented by one rating area in the S/4 HANA and CRM system. You define rating areas in your S/4 HANA system in Customizing for Contract Accounts Receivable and Payable under *Integration* > *Convergent Charging* > *Specifications for Connecting Multiple CC Systems to a FICA System* > *Define Rating Areas*.

In the CRM system using transaction `R3AS`, you then download the rating areas defined in the S/4 HANA system using the corresponding customizing download object `DNL_CUST_RA`. You monitor the current status of the initial load in transaction `R3AM1`.

When connecting multiple SAP CC systems, you must make multiple logical ports available (at least one port for each consumer proxy for each SAP CC system). You assign the logical ports defined in the SOA Manger to your rating areas. In the S/4 HANA system you require logical ports for the following Web services:

- ChargeableItemChargingServices (`CO_FKKCC_CIT_CHARGE_SERVICE`)
- RatingServices (`CO_FKKCC_RATING_SERVICES`)
- RechargingServices (`CO_FKKCC_RECHARGING_SERVICES`)
- RefillingServices (`CO_FKKCC_REFILLING`)
- SUACProvisioningServices (`CO_FKKCC_SUBSCRIBER_ACCOUNT`)

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- AllowanceManagementServices (`CO_ISX_ALLOWANCE_MANAGEMENT_SE`)
- ContractStateManagementServices (`CO_ISX_CC_CONTRACT_STATE_MANAG`)
- ContractProvisioningServices (`CO_ISX_CC_CTR_PROV_SERVICES`)

  This Web service is used for the transfer of provider contracts only.
- CustomerDataExportImportServices (`CO_FKKCC_CUST_DATA_EXP_IMP_SRV`)
- OperationalStatusServices (`CO_FKKCC_OPERATIONAL_STAT_SRV`)

In the S/4 HANA system you assign rating areas to logical ports in Customizing for Contract Accounts Receivable and Payable uner *Integration* > *Convergent Charging* > *Specifications for Connecting Multiple CC Systems to an FICA System*.

In the CRM system you require logical ports for the following Web services:

- ChargeableItemChargingServices (`CO_CRM_ISX_CC_CHARGEABLE_ITEM`)
- RatingServices (`CO_CRM_ISX_CC_RATING_SERVICES`)
- SubscriberMappingTableManagementServices (`CO_CRM_ISX_CC_SUBSCRIBER_MAPPI`)
- SubscriberRangeTableManagementServices (`CO_CRM_ISX_CC_SUBSCRIBER_RANGE`)

In the CRM system you assign logical ports to rating areas in Customizing for Customer Relationship Management under *Cross-Industry Functions* > *Provider Order and Contract Management* > *Joint Distribution and Multiple SAP CC Systems for a SAP CRM System*.

5.2 Contract Transfer Between Multiple SAP CC Systems

If you use multiple SAP CC systems, you can transfer provider contracts from one SAP CC system to another. You trigger the transfer in SAP CRM.

For the transfer SAP CC provides the Web services CustomerDataExportImportServices (`CO_FKKCC_CUST_DATA_EXP_IMP_SRV`) and OperationalStatusServices (`CO_FKKCC_OPERATIONAL_STAT_SRV`) which are consumed by SAP S/4 HANA. To perform the transfer, you have to maintain the corresponding logical ports and rating areas for these Web services in your S/4 HANA system.

> **Note**
>
> Before you transfer provider contracts between SAP CC systems, make sure that the configuration of both systems is identical (for example, regarding the currencies, mapping tables, tax configuration and different plans used).
>
> When you transfer a provider contract, the contract is automatically locked in the source SAP CC system. This prevents the provider contract and its linked contracts from being modified or deleted. The provider contract cannot be used by standard processes such as charging, rerating, refilling, or activation. However, if necessary, you can reactivate a locked contract. After the transfer has been successful, the provider contract receives the status *closed* in the source SAP CC system.

> **Caution**
>
> After having transferred a provider contract:
>
> - You can only perform rerating if you store the source information about the services consumed by the customer as consumption items in the S/4 HANA system.
> - You can recharge a provider contract only as of its transfer date.

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5.3 Reading Counters in SAP Convergent Charging

In SAP CC, counters, among other things, reflect the balance of free units granted to a customer.

You define counters as part of a charge or charge plan in SAP CC and use them during rating and charging to determine prices.

Within a provider contract, counters can be shared, that is, several provider contract items can share one counter. In this case, the technical name and the namespace have to be identical.

You access counters in SAP Convergent Charging using the Web service `ContractStateManagementServices`. Using the operation `chargingContractStateRead`, you can read current counter values.

To display counter values in SAP CRM and SAP S/4 HANA, you have to configure the consumer proxy `CO_ISX_CC_CONTRACT_STATE_MANAG` in the S/4 HANA system accordingly. The CRM system accesses this proxy using the RFC function modules `CRM_ISX_CONTRACT_COUNTER_QUERY` or `CRM_ISX_CONTRACT_READ`and `CRM_ISX_CONTRACT_QUERY`. The parameter `IT_CONTRACT_ID` can pass several contract IDs to the function module `CRM_ISX_CONTRACT_COUNTER_QUERY`. Shared counters are returned in parameter `ET_CONTRACT_COUNTER`, counters of provider contract items are returned in parameter `ET_CONTRACT_ITEM_COUNTER`, and counters of charges in parameter `ET_CONTRACT_ITEM_CHRG_COUNTER`.

If you are using scenario `FICA_CC_1`, the function modules for reading provider contracts, `CRM_ISX_CONTRACT_READ` and `CRM_ISX_CONTRACT_QUERY`, supply the provider contract structure as well as the counter values. Parameter `ET_CHARGING_ITEM_SHAREDCOUNTER` contains shared counters. Parameter `ET_CHARGING_ITEM_NON_SC` contains counters of provider contract items (field `CHARGE` is initial) and of charges (field `CHARGE` is filled).

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6 Transactional Data

The following sections give an overview of the creation and processing of transactional data within the Offer-to-Cash business process.

Order of Creation of Transactional Data

The following figure shows the order in which transactional data is created in SAP CRM, SAP S/4 HANA, and SAP CC.

Example

You first create the billable item class and then the billable item mapping before creating a charged item class.

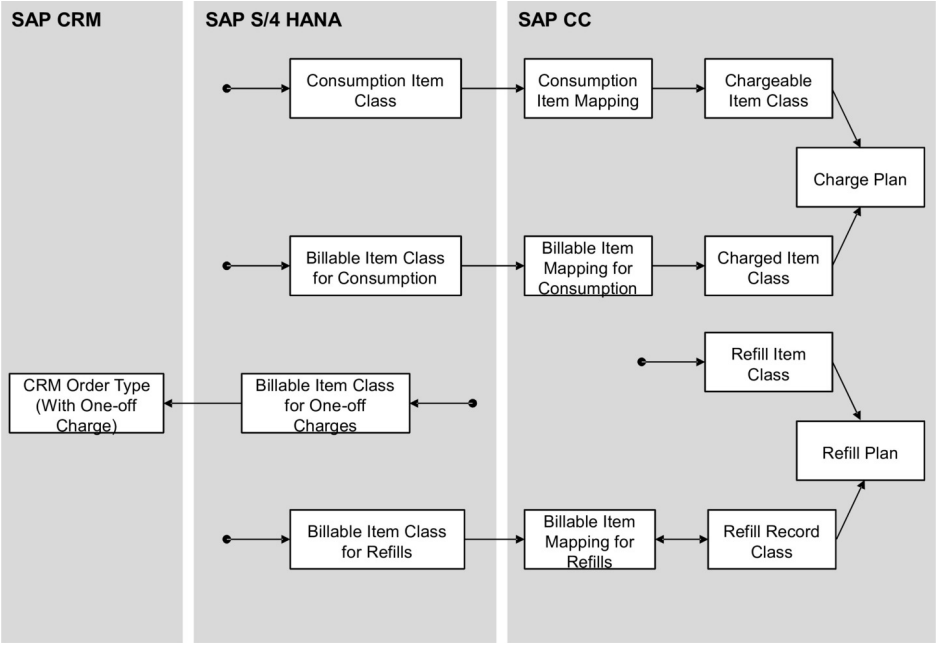


Figure 8: Order of Creation of Transactional Data Within Offer-To-Cash

Relations Between Transactional Data

The following figure shows the relations between the transactional data in SAP CRM, SAP S/4 HANA, and SAP CC.

Example

The billable item mapping references the billable item class and the charged item class.

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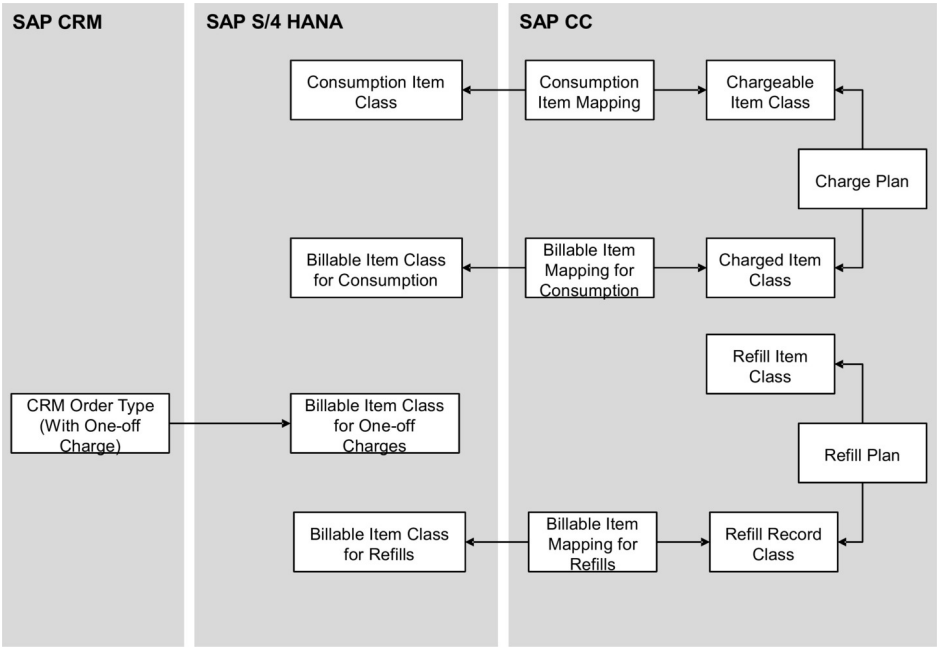


Figure 9: Relations Between Transactional Data Within Offer-To-Cash

Transactional Data in SAP CC

The following figure shows how transactional data is processed in SAP CRM, SAP CI, SAP CC, and between SAP CI and SAP CC.

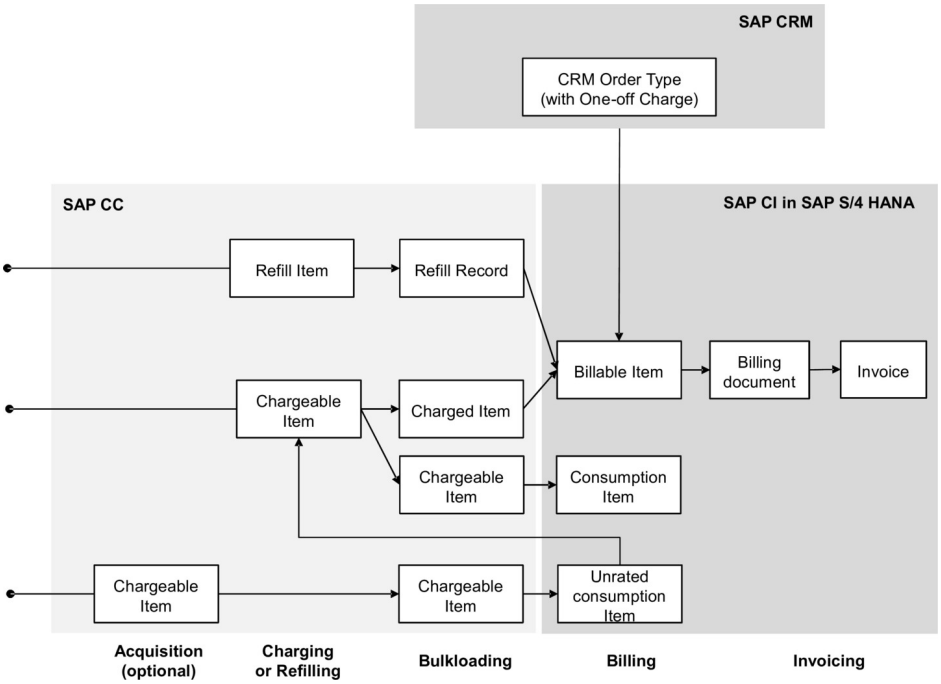


Figure 10: Transactional Data in SAP CC

More Information

For more detailed information see the following sources:

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- For more information on how to integrate Billing with SAP Convergent Charging, see help.sap.com/s4hana > *<Release>* > *Additional Information* > *Product Assistance* > *English* > *Industries* > *Contract Accounts Receivable and Payable* > *SAP Convergent Invoicing*
  - *Billing* > *Configuration of Billable Item Classes* and following
  - *Integration with Other Components and Products* > *Integration with SAP Convergent Charging*
  - *Integration with Other Components and Products* > *Transfer of Billable Items from External Systems* and following

> **Note**
>
> If you are using Offer-to-Cash with another industry solution, in the SAP Help Portal choose the industry solution you are using and then proceed to the documentation for Contract Accounts Receivable and Payable.

- For more information on how to integrate Billing with SAP CRM, see help.sap.com/s4hana > *<Release>* > *Additional Information* > *Product Assistance* > *English* > *Industries* > *Contract Accounts Receivable and Payable* > *SAP Convergent Invoicing* > *Integration with Other Components and Products* > *Integration with SAP Customer Relationship Management* .
- For more information on the interfaces existing between SAP Convergent Charging, SAP CRM and SAP S/4 HANA, see Customizing for Contract Accounts Receivable and Payable under *Integration* > *Convergent Charging* > *> Manage Interfaces to SAP CC* > *Display Overview of Interfaces to SAP CC* .
- For more information on how to integrate the Billing Document and BDR Presentment between SAP Convergent Invoicing and SAP CRM, see help.sap.com/crm704 > *Application Help* > *English* > *SAP CRM for Industries* > *Telecommunications* > *Functions in Interaction Center for Financial Customer Care* > *Bills* .
- For more information on how to integrate Dispute Management in the Interaction Center between SAP Convergent Invoicing and SAP CRM, see help.sap.com/crm704 > *Application Help* > *English* > *SAP CRM for Industries* > *Telecommunications* > *Functions in Interaction Center for Financial Customer Care* > *SAP Dispute Management* .

6.1 Storage of Consumption Data

You store the original consumption information in SAP Convergent Invoicing in the S/4 HANA system. The S/4 HANA system stores the consumption data in the form of consumption items both before and after rating. Storing the consumption data in the S/4 HANA system enables you to:

- Work with scaled prices
- Trigger rerating in SAP Convergent Invoicing
- Use partner settlement
- Correct consumption items in the S/4 HANA system

To implement the **storage of consumption data in SAP Convergent Invoicing** in the S/4 HANA system proceed as follows:

- To set up data storage of consumption items in SAP Convergent Invoicing for a new installation of SAP S/4HANA when SAP Convergent Invoicing is integrated with SAP Convergent Charging, follow these steps:
  1. In SAP Convergent Charging, activate data storage of consumption items in SAP Convergent Invoicing.

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2. In SAP Convergent Invoicing, configure and activate consumption item classes that can receive the data over appropriate interfaces.
3. Replicate the consumption item classes as chargeable item classes and charged item classes to SAP Convergent Charging. Check and configure the mapping of each chargeable item class and each charged item class in SAP Convergent Charging.
4. Configure at least one logical port for each dispatcher provided by SAP Convergent Charging for the Web services *Charging* and *Recharging*.

- If you have already integrated SAP Convergent Invoicing in SAP S/4 HANA with SAP Convergent Charging and you want to move data storage from SAP Convergent Charging to SAP Convergent Invoicing, then proceed as follows:
  1. In SAP Convergent Charging, remove the link between the chargeable item class and the chargeable item package.
  2. In SAP Convergent Charging, activate the synchronization of consumption item classes.
  3. Ensure that the interface component *Link with Consumption Items* (`CIT_REF`) is active in SAP Convergent Invoicing.

     This interface component establishes the link between consumption items and the corresponding billable item class.
  4. Update the mapping of this billable item class in SAP Convergent Charging.
  5. Configure consumption item classes in SAP Convergent Invoicing based on the chargeable item classes used.
  6. In SAP Convergent Charging, synchronize the chargeable item classes with the consumption item classes configured in SAP Convergent Invoicing.
  7. In SAP Convergent Charging, activate data storage of consumption items in SAP Convergent Invoicing.
  8. Enter the date, starting from which you want to store consumption items in SAP Convergent Invoicing in the S/4 HANA system. In Customizing for Contract Accounts Receivable and Payable, choose *Convergent Invoicing* > *Integration* > *SAP Convergent Charging* > *Enter Specifications for the Transition of Data Storage*.
  9. In SAP Convergent Charging, configure the logical ports for the Web services *Charging* and *Recharging*.
  10. In SAP Convergent Invoicing, configure a logical port for the Web services *Charging* and *Recharging* and for the consumer proxy.
  11. Enter the date, starting from which you want to be able to process consumption items from SAP Convergent Invoicing (end of the transition phase).

      In Customizing for Contract Accounts Receivable and Payable, choose *Convergent Invoicing* > *Integration* > *SAP Convergent Charging* > *Enter Specifications for the Transition of Data Storage*.
  12. In SAP Convergent Charging, activate rerating from within SAP Convergent Invoicing.
  13. In SAP Convergent Charging, activate archiving of the charged items still stored in the Batch Acquisition and Rating Toolset (BART).
  14. By relieving the data load on BART, which you achieve by the transition of data storage, you can retain snapshots for counters in SAP Convergent Charging over a longer period of time than before. If needed, extend the residence time for counter snapshots.

6.2 Flow of Consumption Data

From (unrated or rated) chargeable items or refill requests, SAP CC creates charging output data (that is, charged items, refill records, ABM notifications, and unrated chargeable items) and mass data files. SAP CC then

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reads and maps these files to the format of the respective billable item class or consumption item class by calling the respective RFC in the S/4 HANA system. Then SAP CC bulk loads the data as billable items and consumption items into the S/4 HANA system. If you connect multiple FI-CA systems to one SAP Convergent Charging system, SAP CC manages the item storage and transfer for the relevant FI-CA systems based on customer management area information and JCo destinations.

To ensure the completeness of the chargeable and charged items transferred from SAP Convergent Charging to the S/4 HANA system, you should regularly trigger offline charging as well as the calculation of recurring charges. You achieve the latter by regularly taking the appropriate activation steps for provider contracts in SAP Convergent Charging.

The S/4 HANA system establishes the RFC connections (depicted in the second figure above) every time the bulkloader is started in SAP Convergent Charging. This means that the frequency with which you start the bulkloader in SAP Convergent Charging has a direct impact on the data available for further processing in the S/4 HANA system.

More Information

For more information, see section “Billing and Invoicing”.

6.3 Rating of Consumption Data

SAP Convergent Charging receives unrated consumption items from a technical system and enriches them with information related to provider contracts or subscriptions. SAP Convergent Charging then transfers the unrated consumption items to SAP Convergent Invoicing. SAP Convergent Invoicing stores and manages the unrated consumption items, and then sends them back to SAP Convergent Charging for rating.

> **Note**
>
> You can create missing consumption items in SAP Convergent Invoicing manually.
>
> To enhance the input data for rating in SAP Convergent Invoicing, use event 8175.

> **Note**
>
> To support load balancing when you are using multiple logical ports for the *Charging* and *Recharging* Web services, the rating process iterates through all logical ports (responsible dispatchers) for each rating or rerating unit. If a call to a dispatcher fails, the rating process tries to use the next available dispatcher to execute the Web service. The rating process repeats calling the next available dispatcher until either the call succeeds or all ports have returned an error.

More Information

For more information on rating initiated by SAP Convergent Invoicing, see the documentation for SAP Convergent Invoicing under *Data Storage for Consumption Items* > *Rating* > *Points in Time for Rating*.

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6.4 One-Off Charges

One-off charges are transferred from the CRM to the S/4 HANA system automatically using the CRM Middleware (based on the configuration of the order item type). In the S/4 HANA system, one-off charges are processed using a billable item class which is configured for one-off charges using the interface component CRM Order.

For more information on how to transition from rerating based on chargeable items stored in BART to rerating based on consumption items stored in the S/4 HANA system, see service.sap.com/BRIM.

> **Note**
>
> As an alternative, if the calculation of one-off charges is based on the consumption history or if you have to reset technical counters used in the logic during the calculation, you can configure event-based charges in SAP Convergent Charging. For more information, see the Configuration Guide for SAP Convergent Charging at help.sap.com/cc41 > *Configuration and Deployment Information* > *Configuration Guide (SAP Service Marketplace)* > *Install* > *SAP CC 4.1 Configuration Guide* > *Configuration with Integrated Scenario Consume-To-Cash* > *Master Data Basic Settings*.

6.5 Prepaid Data

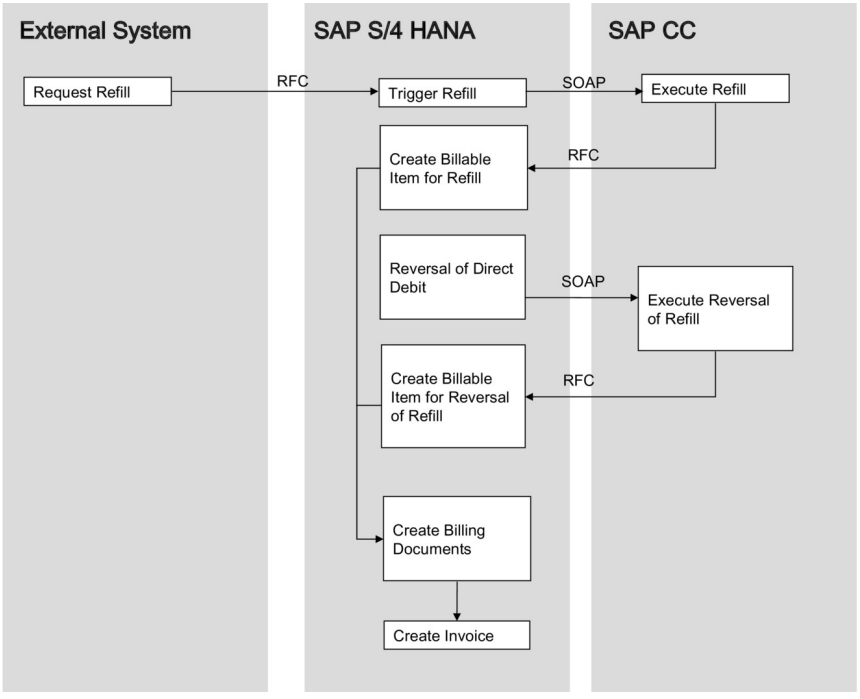


Figure 11: Runtime Integration for Prepaid Data in Offer-to-Cash (Refill)

Billing in SAP Convergent Invoicing receives prepaid data for refills as billable items from SAP Convergent Charging. It stores and manages the data and subsequently groups the data together with existing consumption of the customer and summarizes the data in a billing document. During invoicing, one or more billing documents are merged into an invoice.

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Technical Configuration of Prepaid Data Processing

To enable refills in prepaid scenarios, register the function modules `FKK_CC_SAMPLE_0965`, `FKK_CC_SAMPLE_0966` and `FKK_CC_SAMPLE_0967` for the events 965, 966 and 967, respectively, in the activities in Customizing for Contract Accounts Receivable and Payable under *Prepaid Processing* > *Prepare Integration with SAP Convergent Charging*.

To call the appropriate Web service in SAP Convergent Charging, configure the logical port for the consumer proxy for refills `CO_FKKCC_REFILLING` as described in the section "Setting Up Communication with SAP Convergent Charging Using Web Service".

Refills can either be credits posted to a prepaid account or free units, for example, free minutes or SMS. SAP Convergent Charging stores and manages all refilled units as counters.

To refill amounts or units, you have to create appropriate refill plans in SAP Convergent Charging with the refill logic adapted to the respective scenario.

The refill of amounts can be triggered within SAP Convergent Charging (periodic refill, one-time refill) or by an external source, for example, SAP S/4 HANA. The S/4 HANA system uses the Web service *RefillingServices* for creating refills by means of payment lots or when processing prepaid accounts (transaction `FP40P`). You have to configure the refill logic in SAP Convergent Charging using the usage identifiers `_refill_` and `_reverse_refill_` for the management of refills and the reversal of refills, respectively (see following two figures).

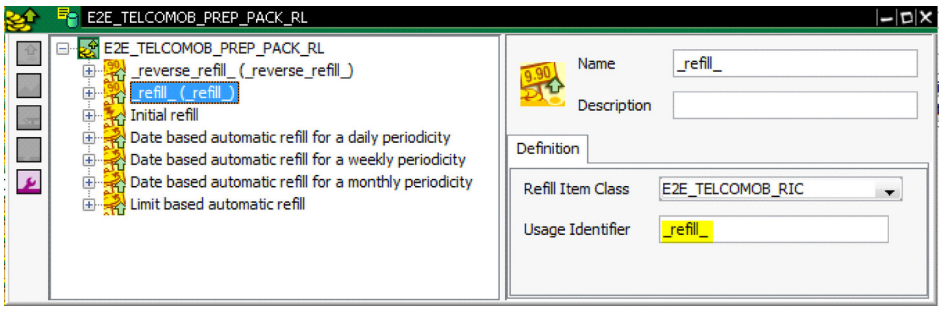


Figure 12: Recurring Component for Refill

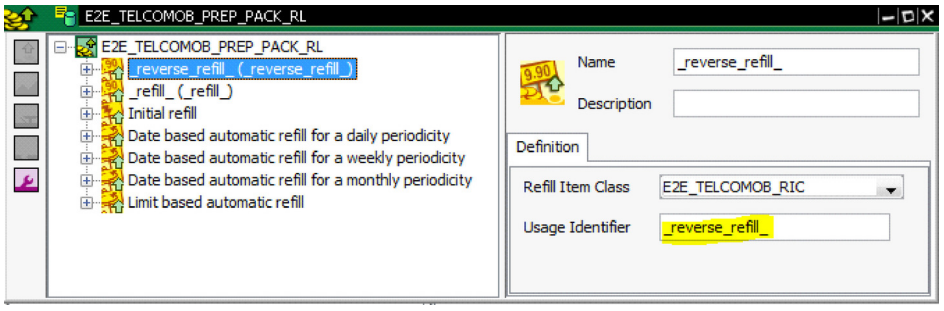


Figure 13: Recurring Component for the Reversal of Refills

Refill plans contain the logic for the generation of billable items (and temporary refill records).

For more information on how to define and maintain refill logic and refill plans, see the Configuration Guide for SAP Convergent Charging at help.sap.com/cc41 > *Configuration and Deployment Information* > *Configuration Guide (SAP Service Marketplace)* > *Install* > *SAP CC 4.1 Configuration Guide* > *Configuration with Integrated Scenario Consume-To-Cash* > *Master Data Basic Settings* > *Setting up the refill logic and a refill plan*.

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Units are refilled using refill packages. You have to maintain refill packages in the S/4 HANA- and CC-systems in parallel. However, in the SAP Convergent Charging system, you only maintain a subset of the attributes of the refill packages using translation tables.

Refill packages can contain up to three types of refillable units (for example, free minutes, free SMS, free amounts of data for downloads) as well as an amount for the prepaid account.

To maintain refill units and packages, proceed as follows:

1. In the S/4 HANA system, define refillable units in Customizing for Contract Accounts Receivable and Payable under *Prepaid Processing* > *Refills* > *Define Refillable Units*.
2. Define refill packages in the S/4 HANA system in Customizing for Contract Accounts Receivable and Payable under *Prepaid Processing* > *Refills* > *Define Refill Packages* (see following two figures).

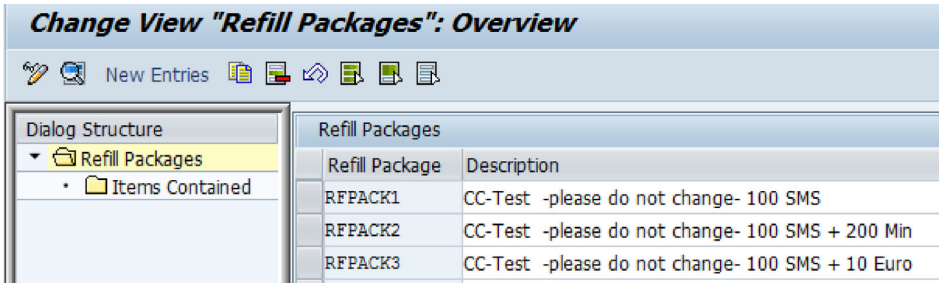


Figure 14: Refill Package in Customizing for Contract Accounts Receivable and Payable

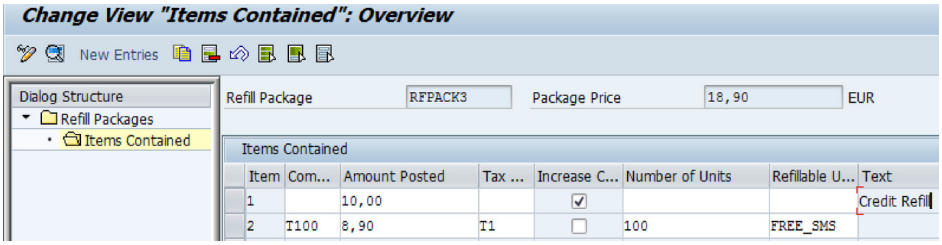


Figure 15: Items of a Refill Package in Customizing for Contract Accounts Receivable and Payable

3. Enter these refill packages in SAP Convergent Charging in the refill plan in the form of a translation table (see figure below).

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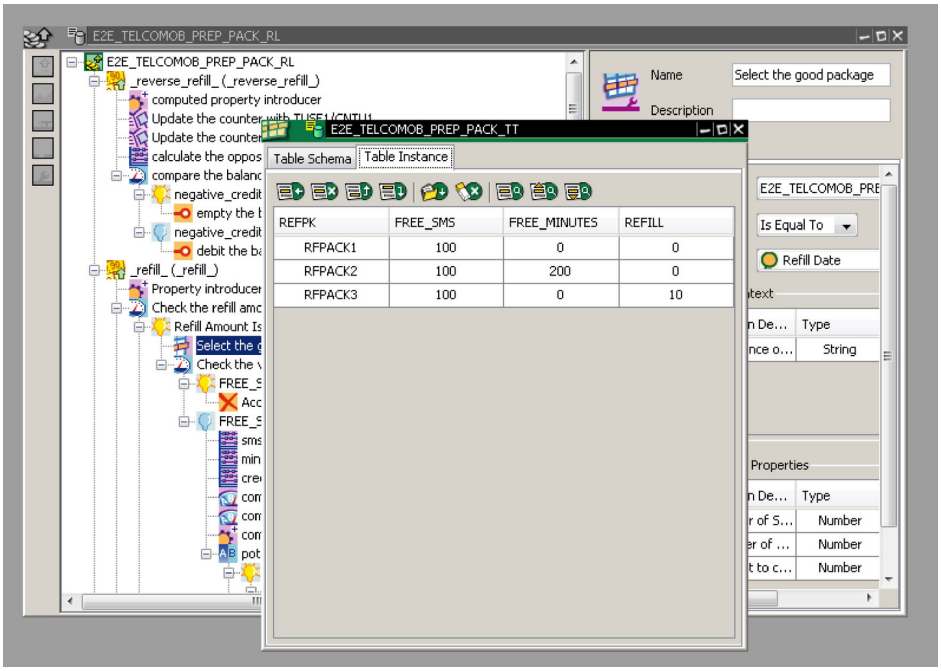


Figure 16: Configuration of a Refill Package in SAP Convergent Charging

Make sure that your entries in the translation table correspond to the entries you made in S/4 HANA Customizing.

6.6 Billing and Invoicing

Transactional data in the form of billable items originates in SAP Convergent Charging as consumption and refill data and in SAP CRM as one-off charges. SAP Convergent Charging triggers periodic or event-based charges and creates billable items in the S/4 HANA system.

Billing receives billable items, which represent charged consumption and refill data from SAP Convergent Charging, and charges from SAP CRM (and SAP Convergent Charging). It stores and manages the data and subsequently groups the data together with other existing consumption of the customer and summarizes the data in a billing document. This means that billing documents pre-aggregate billable items. During invoicing, one or more billing documents can then be merged into one invoice.

Based on the billing document and possibly on additional source documents from other systems, Invoicing creates an invoice for the customer and posts this total amount to the contract account of the business partner as a receivable.

You manage the receivable using the standard processes of Contract Accounts Receivable and Payable and collect the receivable from the customer (incoming payment, dunning).

To enable the flow of consumption and refill data between the S/4 HANA system and SAP Convergent Charging, the following steps are required:

- As a general step for both consumption and refill data, synchronize the currencies used in the S/4 HANA and SAP Convergent Charging systems.
- For consumption data carry out the following steps:

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1. Define consumption item classes in the S/4 HANA system so that the consumption items correspond to the chargeable items sent by the external mediation system for the usage of a service (consumption). You can add fields to the fields used by the external mediation system to use in the charge.
2. Define chargeable item classes and the respective consumption item mapping in SAP Convergent Charging, referencing the consumption item classes created in the S/4 HANA system (see first figure above).
3. Define billable item classes in the S/4 HANA system.

   In order to enable the drill-down from billable items to their respective consumption items, activate the interface components *SAP Convergent Charging* and *Link to consumption item*.
4. Define charged item classes and the respective billable item mapping in SAP Convergent Charging, referencing the billable item classes created in the S/4 HANA system (see first figure above).
5. Set up the generation of charged items in the charge plans.

- For refill data carry out the following steps:
  1. Define a billable item class for refills in the S/4 HANA system.
  2. Define refill item classes in SAP Convergent Charging that correspond to the refill items sent by the S/4 HANA system.
  3. Define refill record classes and the respective billable item mapping in SAP Convergent Charging, referencing the billable item class created in the S/4 HANA system (see second figure above).
  4. Set up the generation of refill records in the refill plans.

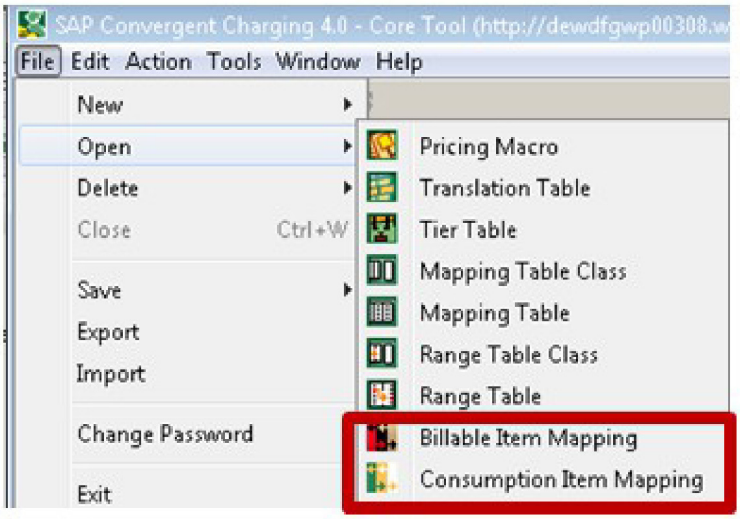


Figure 17: Menu Entry of the Billable and Consumption Item Mapping in the SAP Convergent Charging System

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7 Contract Data

Offer to Cash works with the following types of contracts and agreements:

- Provider contract

  A provider contract comprises all legally binding agreements regarding the provision and billing of services that are entered into by a customer and a company for a specified period of time.

  If you have made the appropriate system settings, the master data of provider contracts is automatically replicated from the CRM system to the S/4 HANA system. For the integration and data distribution of the provider contract in SAP S/4 HANA and SAP CC, you can use the RFC `CRM_ISX_CONTRACT_MAINTAIN`.

- Master agreement

  A master agreement represents a contractual agreement between a service provider and a business customer.

  In a master agreement, you specify the fundamental elements and conditions for dependent individual contracts (provider contracts). In addition, you can agree on rules in the master agreement that govern the business activity that arises from these individual contracts.

  The master agreement can contain agreements on products, prices, discounts, and the procedure for invoice creation. It also contains the additional parties (represented by business partners), who can sign individual contracts under the conditions of the master agreement.

  In SAP CRM, you maintain those components that make it easier to conclude individual contracts, such as the products to which the conditions of the master agreement apply, the parties (business partners) that are authorized to sign individual contracts that refer to the master agreement, and rules that specify which business partners are allowed to purchase which products.

  The elements that are needed in SAP S/4 HANA are replicated automatically to the S/4 HANA system, that is, the list of products and the business partners that are included in the master agreement.

  In the S/4 HANA system, you add entries to the master agreement that are needed for the processes for invoice creation or for subsequent discounting. You make these entries in the form of the following agreements:

  - Agreement for creation of invoices (invoice agreement)
  - Agreement for discounting (discount agreement)
  - Agreement for creation of invoicing lists

- Partner agreement

  Partner agreements are used to share revenue with a content provider or to pay royalties. With end customers you have provider contracts, with partners you have partner agreements.

  In technical terms, partner agreements are provider contracts of the contract category *Partner Agreement* (indicated by an entry in the *Contract Specification* field). You create partner agreements in the CRM system and replicate them to the S/4 HANA system. Partner agreements are distributed between the CRM and S/4 HANA systems, as well as the SAP CC system, in the same way as provider contracts.

  In SAP CRM, you can create partner agreement orders and partner agreements. Also, you can define products as partner products that can be used in partner agreements.

- Sharing contract

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A sharing contract specifies services, as for example, buckets of free minutes, SMS, or data, that apply to all individual contracts assigned to a sharing contract and which can be used by all contract partners.



Example

A corporate customer with several business units signs a master agreement with a communications service provider. All employees of all business units receive collectively 1000 free minutes every month. The shared free minutes are consumed according to the first-come first-served principle.

Sharing contracts are provider contracts that are specified by the contract category *Sharing Contract* in the CRM system and by the contract specification *Sharing Contract* in the S/4 HANA system. Provider contracts created with reference to such a counter sharing contract can consume from the shared services (buckets). You create a sharing contract at item level in the master agreement in SAP CRM. In that case provider contracts created with reference to this master agreement item are automatically linked to the counter sharing contract.

Counter sharing contracts can be relevant beyond counter sharing, for example, they can have their own recurring charges.

More Information

For more information, see the documentation of:

- SAP Convergent Invoicing under “Master Data in SAP Convergent Invoicing”
- SAP Customer Relationship Management under help.sap.com/crm704 › *SAP CRM for Industries* › *Telecommunications* › *Functions for Provider Sales and Order Management* › *Partner Agreement* › and *Sharing Contract*

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8 Discounts and Charges on Billable Items

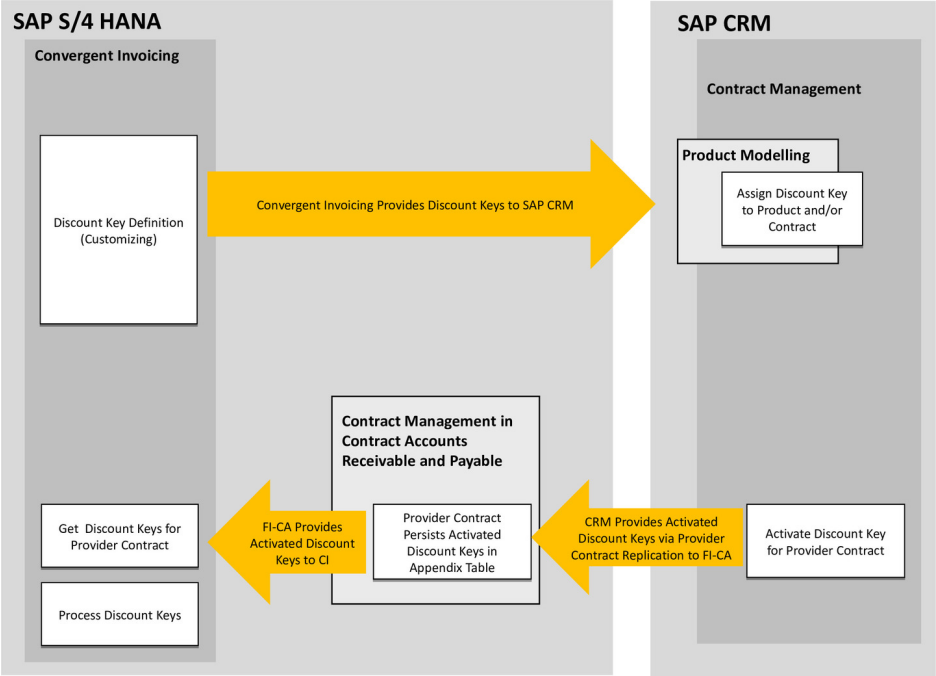


Figure 18: Integration of Discounts and Charges on Billable Items (with SAP CRM and SAP S/4 HANA)

The figure above illustrates the process of configuring and calculating discounts and charges in a business-to-consumer scenario as follows:

- You define discounts and charges in the S/4 HANA system.
- Discount and charge keys are replicated to the CRM system. You assign discount or charge keys during product modeling and they are activated when a contract is signed.
- The discount key assignment is transferred to the S/4 HANA system when the provider contract is replicated and it is stored as an appendix to the provider contract in Contract Accounts Receivable and Payable (FI-CA).
- SAP Convergent Invoicing (CI) applies and processes discounts and charges.

Technical Configuration

S/4 HANA system

In order to calculate discounts and charges, you have to make the appropriate settings in the S/4 HANA system in Customizing for Contract Accounts Receivable and Payable under *Convergent Invoicing* > *Billing* > *Additional Functions* > *Discounts and Charges on Billable Items*.

SAP S/4 HANA provides discount keys and discount texts using the RFC function module `FKK_BIX_DISCOUNT_GET_API`, which is also called during replication of discount keys to the CRM system.

CRM system

You initiate the replication of discount keys to the CRM system via initial load using transaction `R3AS` and the load object `DNL_CUST_DISCO`in the CRM system. You can view the result in the CRM system in Customizing for Customer Relationship Management under *Cross-Industry Functions* > *Provider Order and Contract*

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Management > *Transactions* > *Billing Discounts* > *Display Billing Discounts* . After they have been replicated, you can assign the discount keys to products and use them during the processing of sales orders.

The CRM system replicates the discount assignment to the provider contract when the provider contract is replicated in the S/4 HANA system using the order distribution infrastructure (ODI). ODI calls the RFC `CRM_ISX_CONTRACT_MAINTAIN`, which ensures that the provider contract and discount assignment are created in the S/4 HANA system.

You can verify whether a discount key has correctly been assigned to a provider contract in the S/4 HANA system by displaying the provider contract and its respective contract items in the S/4 HANA system using transaction `FP_VT3`. The appendix table `DFKK_VT_D` contains the assigned discount keys.

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9 Partner Settlement

If an end user uses services or purchases products of a service provider, and these services or products use rights, infrastructure, services, or products of a partner, then payment is due to this partner. In these cases, a portion of the amount that you invoice to the end customer has to be paid to one or more partners.

The overall solution for such business transactions is called Revenue Sharing and Partner Settlement or simply Partner Settlement.

If you want to use partner settlement, it is a prerequisite to use the data storage of consumption items in SAP Convergent Invoicing (see section “Transactional Data Within the Offer-to-Cash Business Process”).

The basis for partner settlement is the customer billable items and the partner billable items linked to them. The customer billable items are viewed as primary items and partner items as secondary items; partner billable items contain a reference to the customer billable items that they belong to.

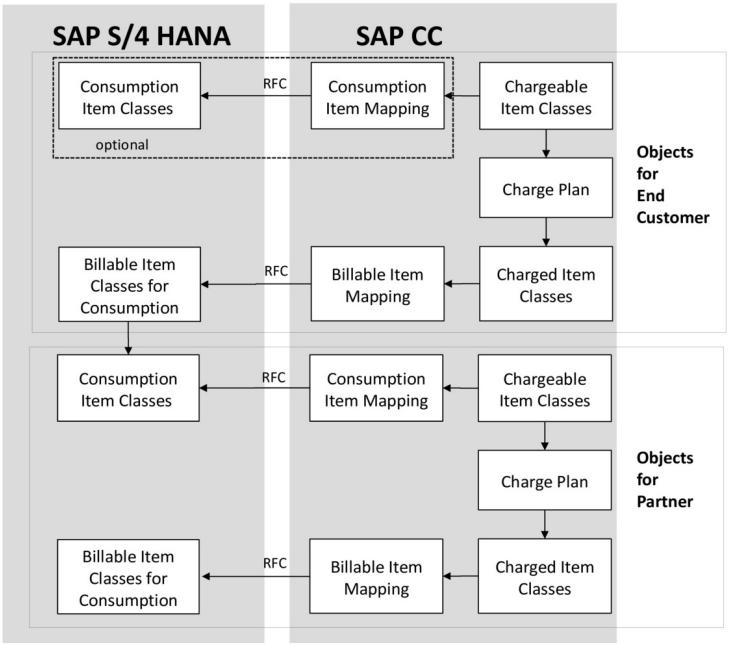


Figure 19: Partner Settlement

To be able to use partner settlement, the following configuration, and technical setup steps are required (see figure above):

1. Define a billable item class for the customer billable items in SAP Convergent Invoicing and activate the interface component *Partner Settlement: Basic Data of Customer Item* (`CUST_ITEM`). This automatically activates the prerequisite interface component *Primary Items* (`PRIMARY_IT`).
2. Define the billable item mapping in SAP Convergent Charging, referencing this billable item class.
3. **Optional**

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Define a consumption item class for the partner consumption items in SAP Convergent Invoicing, as well as a chargeable item class and the consumption item mapping in SAP Convergent Charging, referencing this consumption item class.

Based on the customer billable items, the dependent partner consumption items are created in SAP Convergent Invoicing.

4. Define a consumption item class for these partner consumption items in SAP Convergent Invoicing and activate the interface component *Partner Settlement: Basic Data of Partner Item* (`PARTNER_ITEM`). This automatically activates the prerequisite interface component *Secondary Item* (`SECONDARY_IT`). Using this component, the fields for the link information to the customer billable item are included in the partner consumption item(s).
5. Define a chargeable item class and the consumption item mapping in SAP Convergent Charging, referencing this consumption item class.

   Using data storage in SAP Convergent Invoicing, partner consumption items are sent to SAP Convergent Charging for rating. The partner billable items are then sent from SAP Convergent Charging to SAP Convergent Invoicing.
6. Define a billable item class for partner billable items in SAP Convergent Invoicing and activate the interface component *Partner Settlement: Basic Data of Partner Item* (`PARTNER_ITEM`).
7. Define a billable item mapping in SAP Convergent Charging, referencing this billable item class.
8. In order for the the partner billable item (partner charged item in SAP Convergent Charging) to inherit the link information from the partner consumption item (partner chargeable item in SAP Convergent Charging), you need to map the following fields from the chargeable item to the same fields of the charged item in the charge plan in SAP Convergent Charging:
   - `GPART_PRIMARY`
   - `VTREF_PRIMARY`
   - `SUBAP_PRIMARY`
   - `SRCTAID_PRIMARY`
   - `SRCTATY_PRIMARY`
   - `RELTYPE_PRIMARY`

Example

See the mapping for the field `GPART_PRIMARY` below:

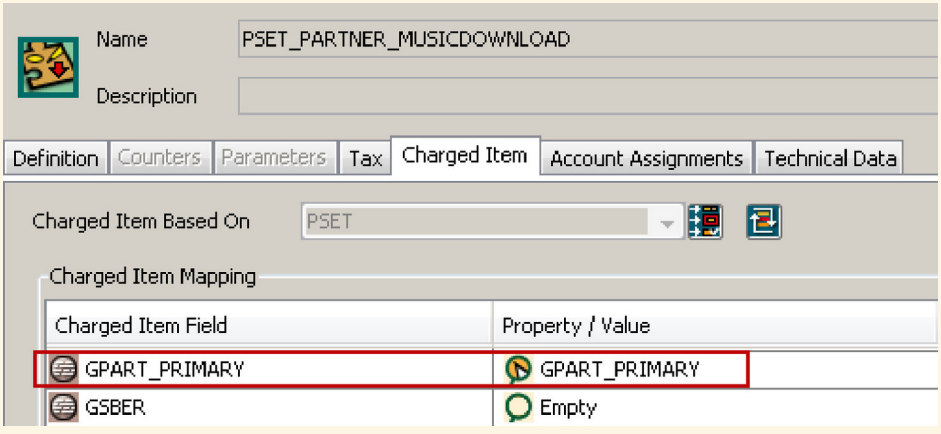


Figure 20: Mapping for the GPART_PRIMARY Field

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10 Allowances Within the Offer-to-Cash Business Process

You configure allowances in SAP CC as follows:

1. Create the predefined chargeable item class ALLO in SAP CC.
2. Customize the allowance interface in SAP CC.
3. Create an allowance definition table class in SAP CC.
4. Create an allowance definition table in SAP CC.
5. Configure the allowance creation in charge plans in SAP CC.

You maintain allowances in products in SAP CRM.

To comply with legal requirements you can post deferred revenue for allowances.

For more information see the following sections:

10.1 Creation of the Predefined ALLO Chargeable Item Class in SAP CC

SAP CRM creates allowances in SAP CC by sending charging requests for chargeable items. The chargeable item contains the identifiers of the allowance definition used to initialize the allowance. As a prerequisite, the predefined ALLO chargeable item class must exist in SAP CC and it must contain the predefined fields ALLOW_DEF_ID (identifier of the allowance definition used to create the allowance) and ALLOW_START_DATE (date on which the allowance must be activated).

The charges must use this chargeable item for creating the allowances from the allowance definition ID.

The created allowance must be initialized from the specified allowance definition and must be active on the specified date.

You must create the corresponding ALLO consumption item class in SAP CI to enable the loading of the consumption items into SAP CI. SAP CI provides the corresponding interface component ALLO_CREATE which also contains the fields ALLOW_DEF_ID and ALLOW_START_DATE.

The Core Tool provides you with the ALLO chargeable item template to create a chargeable item in SAP CC. Proceed as follows:

1. Open the ALLO_chargeable_item_class.xml XML file located in the installation directory of your Core Tool in the templates folder.
2. Replace ${catalog} with the name of your catalog.
3. Save and close the XML file.
4. Log on to the Core Tool of SAP CC.
5. Import the chargeable item class.

   For more information, see the documentation of the Core Tool.
6. Create the ALLO consumption item class in SAP CI (optional step).

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7. Create the corresponding consumption item mapping in the SAP CC Core Tool (optional step).

> **Note**
> Perform step 6 and step 7 only if you store consumption data in SAP CI.

10.2 Customizing of the Allowance Interface in SAP CC

To be able to display allowances and allowance plans in SAP CRM, they must have a predefined parameter of type *String* named `ALLOW_DEF_ID`. As a main parameter of the allowances, `ALLOW_DEF_ID` must be initialized when the allowance is created and must be used to customize the allowance.

To include the `ALLOW_DEF_ID` parameter in allowances, proceed as follows:

1. Log on to the Core Tool of SAP CC.
2. In the menu, choose *Tools* > *Allowance Interface*.
3. Add a new line and then set the following column values:

   Table 5

Column	Value
Name	`ALLOW_DEF_ID`
Type	*String*
Description	The ID of the allowance definition from which the allowance has been initialized.
Default Value	NotSet

4. In the menu, choose *File* > *Save* to save the allowance interface.

10.3 Creation of an Allowance Definition Table Class in SAP CC

Procedure

Allowance Definitions are managed as mapping tables stored in SAP CC. The structure of the allowance definition tables is defined by using mapping table classes that must contain the following:

- An input column of type *String* named `ALLOW_DEF_ID`

  This column must contain the identifier of the allowance definition.
- An output column of type *String* named `ALLOW_DEF_DESC`
- Additional info of type *String* named `SAP_DATA_TYPE` set to *AllowanceDefinition*

The following table shows an example of an allowance definition table class:

Table 6

Column	Name	Type	Description

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Input Column 1	`ALLOW_DEF_ID`	*String*	The identifier of the allowance definition
Output Column 1	`ALLOW_DEF_DESC`	*String*	A human readable short description of the allowance definition
Output Column 2	`BucketQuantity`	*Number*	The quantity of the allowance
Output Column 3	`Validity`	*Number*	The validity period in days of the allowance

This example shows that the purpose of an allowance definition is to define the quantity and the validity period of an allowance limited in quantity and in time.

The Core Tool provides you with a template to create an allowance definition table class in SAP CC. Proceed as follows:

1. Open the `allowance_definition_table_class.xml XML` file located in the installation directory of the Core Tool in the templates folder.
2. Replace `${catalog}` with the name of your catalog.
3. Save and close the XML file.
4. Log on to the Core Tool of SAP CC.
5. Open the mapping table class from the XML file.

   For more information, see the documentation of the Core Tool.
6. Rename the displayed mapping table.
7. Enter a description for the mapping table.
8. Add input and output columns.
9. Save the mapping table class.

10.4 Creation of an Allowance Definition Table in SAP CC

Allowance definitions are stored in the allowance definition table that is a mapping table which contains:

- An allowance definition table class
- Additional info of type *String* named `SAP_DATA_TYPE` set to *AllowanceDefinition*

Note

Before you can use an allowance definition table in SAP CRM, you must create it in SAP CC.

Example

The following table shows an example of an allowance definition table:

Table 7

ALLOW_DEF_ID	Start	End	BucketQuantity	Validity
A01	201x	201x	100	30

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A02	**201x**	**201x**	**600**	**60**
A03	**201x**	**201x**	**1000**	**60**

A01 defines a credit allowance limited in quantity to 100 units and in time for a period of 30 days. The allowance definition is valid for a period of one year.

The Core Tool provides you with a template to create an allowance definition table in SAP CC. Proceed as follows:

1. Open the allowance_definition_table.xml XML file located in the installation directory of the Core Tool in the templates folder.
2. Replace ${catalog} with the name of your catalog.
3. Replace ${mappingTableClass} with the identifier of an allowance definition table class.
4. Save and close the XML file.
5. Log on to the Core Tool of SAP CC.
6. Open the mapping table from the XML file.

   For more information, see the documentation of the Core Tool.
7. Rename the displayed mapping table.
8. Add allowance definitions as mapping table rows if necessary.
9. Save the mapping table class.

10.5 Configuration of the Allowance Creation Plans in SAP CC

To create an allowance from the allowance definition, you must first design the corresponding charge that creates an allowance in response to the reception of an ALLO chargeable item. You then add the charge to charge plans as a charge plan item. The charge plan item can have several user service identifiers, but one user service identifier must match the following pattern so that the SAP CRM system is able to identify which user service identifier must be used to activate the charge creating the allowance from the allowance definition:

Table 8

ALLOW_* where * stands for 0 to 6 characters. For instance, ALLOW_ and ALLOW_TUT are valid user service identifiers.

Note

The charge plan item must have one and only one user service identifier matching the above pattern. The charge plan must have at least one mapping table ID of type *parameter* initialized with a default allowance definition table instance.

Refer to the documentation of the Core Tool to model the charge and the charge plans.

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10.6 Maintaining Allowances in SAP CRM Products

Procedure

As allowance definition details are not replicated from SAP CC to SAP CRM, most actions regarding allowances in SAP CRM require an active connection to SAP CC. During the following actions, allowance information is requested from or sent to SAP CC:

- During the maintenance of allowance definition groups when:
  - Selecting an allowance definition table ID
  - Adding allowance definitions to a group
  - Displaying allowance definition details
- During product maintenance when displaying the content of the allowance definition table
- During order and contract maintenance when:
  - Displaying allowance definition details of allowances included in a provider order
  - Automatically activating allowances after contract creation or contract change processes
  - While executing the contract process *Allowance Refill* to display allowance definition details and activate allowances
- When displaying allowance details for active contracts

The following web service proxies are used during communication with SAP CC:

- `CO_CRM_ISX_CC_CATALOG_SERVICES`
- `CO_CRM_ISX_CC_CHARGEABLE_ITEM`
- `CO_ISX_ALLOWANCE_MANAGEMENT_SE`

  This proxy resides in SAP S/4 HANA and is called using an RFC.

These web service proxies require valid logical port customizing. Also, you have to customize the order distribution infrastructure to perform the step *Call allowance activation* (`ALLW`). The implementation delivered for this step calls the corresponding web service in SAP CC. Proceed as follows:

1. Configure the logical ports for web service calls in SAP CRM and SAP S/4 HANA according to the instructions given in section Setting Up Communication with SAP Convergent Charging Using Web Services [page 49].
2. Configure the order distribution infrastructure.
3. Maintain allowance definition groups and allowance assignments to products.

   For more information see the documentation of SAP CRM under *Functions for Provider Sales and Order Management* > *Provider Order and Provider Contracts* > *Allowances*.

10.7 Setting up Allowances for the Posting of Deferred Revenues

In order to post deferred revenues in Contract Accounts Receivable and Payable (FI-CA), allowances should be modeled to generate a billable item when:

- An allowance is created, to report its initial quantity and the corresponding revenue to be recognized
- An allowance is consumed, to report the consumed quantity

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- An allowance expires, to report the remaining quantity

To configure the master data:

1. **Create a Billable Item Class for Deferred Revenue**

   Create a billable item class that generates billable items for allowances in SAP CI and activate the following interface components for this billable item class. Then create the billable item mapping in SAP CC.

   Table 9

Interface Component	Field Name	Field Description	Possible Values	Meaning in the Allowance Context
ALLOWANCE	ALLOWANCE	Allowance	' ' indicates that the billable item does not represent an allowance. *X* indicates that the billable item represents an allowance.	Indicates whether a billable item is created within the allowance context
DEF_REV_BY_E (event-based deferred revenue)	DEFREV_CAT	Category of deferred revenues	01: Time based deferred revenues 02: Event-based deferred revenues	This example shows event-based deferred revenue; so the value is 02.
	DEFREV_PDATE	Transfer posting date for deferred revenues	Date	Date of the allowance creation or allowance usage
	DEFREV_ACTION	Action code for deferred revenues	01: Create deferred revenue 02: Recognize revenue	01 indicates that the billable item represents a newly created allowance. 02 indicates that the billable item represents usage of an allowance.
	DEFREV_CTYPE	Cost type or revenue type	Table TFK157	Determines the account (see posting area 1222) and the deferred revenue calculation base (amount or quantity).
	DEFREV_ASSKY	Assignment key	n/a	Allowance ID
QUANTITY	BIT_QUANTITY	Billing quantity of billable item	n/a	The display differs depending on the value set for DEFREV_ACTION. A quantity in combination with the action *01* triggers the posting of deferred revenue. A quantity in combination

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				with the action *02* triggers the recognition of revenue if the calculation base is “quantity”(see DEFREV_CTYPE).
BILL_BASIC	BIT_AMOUNT	Amount of billable item	n/a	The display differs depending on the value set for DEFREV_ACTION.An amount in combination with the action *01* triggers the posting of deferred revenue. An amount in combination with the action *02* triggers the recognition of revenue if the calculation base is “amount” (see DEFREV_CTYPE).
	BIT_CURR	Currency of billable item		

2. **Model allowances to trigger the posting of deferred revenue**

   Billable items generated to trigger the posting of deferred revenue require specific information from SAP CC. Deferred revenue is only relevant for allowances when the revenue corresponding to the consumed or lost credit can be recognized. An allowance that generates a billable item that triggers the posting of deferred revenue must feature the following:

   - A trigger the system executes when the allowance is created

     This trigger generates the following billable item properties:

     Table 10

Interface Component	Field Name	Value
ALLOWANCE	ALLOWANCE	*X*
DEF_REV_BY_E (event-based deferred Revenue)	DEFREV_CAT	*2*
	DEFREV_PDATE	The event date
	DEFREV_ACTION	*1*
	DEFREV_CTYPE	A custom value
	DEFREV_ASSKY	The allowance ID
QUANTITY	BIT_QUANTITY	The initial quantity of the allowance
	BIT_QTY_UNIT	
BILL_BASIC	BIT_AMOUNT	The cost of the allowance
	BIT_CURR	

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- Event-based triggers for debiting the allowance quantity

  These triggers must be enhanced to generate the following billable item properties:

  Table 11

Interface Component	Field Name	Value
ALLOWANCE	ALLOWANCE	*X*
DEF_REV_BY_E (event-based deferred Revenue)	DEFREV_CAT	*2*
	DEFREV_PDATE	The event date
	DEFREV_ACTION	*2*
	DEFREV_CTYPE	A custom value
	DEFREV_ASSKY	The allowance ID
QUANTITY	BIT_QUANTITY	The consumed quantity of the allowance
	BIT_QTY_UNIT	
BILL_BASIC	BIT_AMOUNT	*0*
	BIT_CURR	

- A trigger the system executes when the allowance is expired

  This trigger generates the following billable item properties:

  Table 12

Interface Component	Field Name	Value
ALLOWANCE	ALLOWANCE	*X*
DEF_REV_BY_E (event-based deferred Revenue)	DEFREV_CAT	*2*
	DEFREV_PDATE	The event date
	DEFREV_ACTION	*2*
	DEFREV_CTYPE	A custom value
	DEFREV_ASSKY	The allowance ID
QUANTITY	BIT_QUANTITY	The remaining quantity of the allowance
	BIT_QTY_UNIT	
BILL_BASIC	BIT_AMOUNT	*0*
	BIT_CURR	

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11 Dispute Management

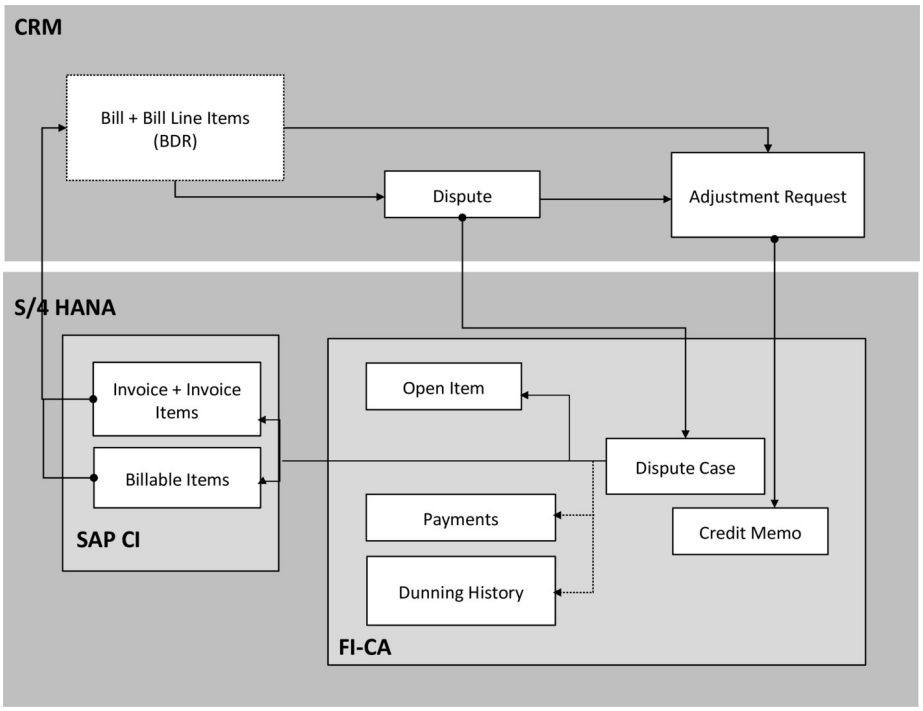


Figure 21: Integration of Dispute Management (with SAP CRM and Convergent Invoicing in FI-CA)

Billing detail records and bills are read from Convergent Invoicing and displayed in SAP CRM. They are, however, not persisted in SAP CRM. From the bill, you create a dispute. The dispute is replicated to Contract Accounts Receivable and Payable (FI-CA) to an FI-CA dispute case. The FI-CA dispute case contains information on the receivable related to the bill and the disputed object.

The dispute is the central unit (a folder) used to process and to track dispute processing. You can create disputes in relation to bills, bill line items, or billing detail records. Disputes contain various attributes as well as notes used for processing and tracking purposes. In addition, all objects and documents used for processing of a dispute can be linked to the dispute, no matter whether they are physically saved in the CRM or the S/4 HANA system (Contract Accounts Receivables and Payables (FI-CA)).

You can create an adjustment request from a bill or as a subitem of a dispute. You can post adjustment requests directly in Contract Accounts Receivables and Payable (FI-CA) as a credit item. In this way, you can access bills from a non-SAP rating and billing system without persisting them in the CRM system.

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12 Setting Up Communication with SAP Convergent Charging Using Web Services

To call a SAP CC Web service from SAP S/4 HANA or SAP CRM, you configure a logical port for each corresponding consumer proxy in SAP S/4 HANA or SAP CRM.

To guarantee availability and enable load balancing during rating, define one port per SAP CC dispatcher instance for all services used for rating (see column “Enabled for Use with Multiple Dispatchers”).

The following consumer proxies are provided:

- Between SAP CRM and SAP CC (consumer proxies on SAP CRM side):

  Table 13

Internal Name of Consumer Proxy	Corresponding SAP CC Web Service	Enabled for Use with Multiple Dispatchers
CO_CRM_ISX_CC_CATALOG_SERVICES	Catalog Management	
CO_CRM_ISX_CC_RATING_SERVICES	Business Process Management	
CO_CRM_ISX_CC_SUBSCRIBER_MAPPI	Subscriber Account Mapping Table Management	
CO_CRM_ISX_CC_SUBSCRIBER_RANGE	Subscriber Account Range Table Management	
CO_CRM_ISX_CC_CHARGEABLE_ITEM	Charging	X

- Between SAP S/4 HANA and SAP CC (consumer proxies on SAP S/4 HANA side):

  Table 14

Consumer Proxy Internal Name	Corresponding SAP CC Web Service	Enabled for Use with Multiple Dispatchers
CO_FKKCC_CIT_CHARGE_SERVICE	Charging	X
CO_FKKCC_PREPAID_ACCOUNT_STATE	Prepaid AccountState Management	
CO_FKKCC_RATING_SERVICES	Business Process Management	
CO_FKKCC_RECHARGING_SERVICES	Recharging	X
CO_FKKCC_REFILLING	Refilling	
CO_FKKCC_SUBSCRIBER_ACCOUNT	Subscriber Account Management	
CO_ISX_CC_CONTRACT_STATE_MANAG	Charging ContractState Management	
CO_ISX_CC_CTR_PROV_SERVICES	Charging Contract Management	

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CO_ISX_ALLOWANCE_MANAGEMENT_SE	Allowance Management	X

The following table gives templates for the URL of the main Web services published by SAP CC. To configure your logical ports enter the URL as indicated in the table:

Table 15

Corresponding SAP CC Web Service	URL templates
Catalog Management	http[s]://<UPDATER_HOST>:<UPDATER_WS_PORT>/catalog?wsdl
Charging	http[s]://<DISPATCHER_HOST>:<DISPATCHER_WS_PORT>/chargeableItemCharging?wsdl
Prepaid Account State Management	http[s]://<UPDATER_HOST>:<UPDATER_WS_PORT>/prepaidaccountstate?wsdl
Business Process Management	http[s]://<UPDATER_HOST>:<UPDATER_WS_PORT>/rating?wsdl
Recharging	http[s]://<DISPATCHER_HOST>:<DISPATCHER_WS_PORT>/recharging?wsdl
Refilling	http[s]://<UPDATER_HOST>:<UPDATER_WS_PORT>/v1/refilling?wsdl
Subscriber Account Management	http[s]://<UPDATER_HOST>:<UPDATER_WS_PORT>/suacProvisioning?wsdl
Charging Contract State Management	http[s]://<UPDATER_HOST>:<UPDATER_WS_PORT>/contractstate?wsdl
Charging Contract Management	http[s]://<UPDATER_HOST>:<UPDATER_WS_PORT>/contractProvisioning?wsdl
Allowance Management	http[s]://<DISPATCHER_HOST>:<DISPATCHER_WS_PORT>/allowanceManagement?wsdl

Note

Enter `https:` or `http:` depending on whether SAP CC publishes the respective Web service using SSL.

`UPDATER` indicates that you have to enter the respective `UPDATER` instance.

`DISPATCHER` indicates that you have to enter the respective `DISPATCHER` instance.

Note

If Secure Sockets Layer (SSL) has been enabled for the SAP CC Web services, the certificate of SAP CC must be trusted by SAP S/4 HANA.

If the SSL dual authentication has been enabled in SAP CC, you require a certificate for SAP S/4 HANA.

You must install the SAP S/4 HANA key pair in the S/4 HANA system and the S/4 HANA certificate must be trusted by SAP CC.

For more information, see the SAPNetWeaver Security Guide under *Network and Communication Security* > *Transport Layer Security*.

To configure logical ports proceed as follows:

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1. Call transaction `SOAMANAGER` in the CRM or S/4 HANA system (depending on the consumer proxy; see list above).
2. Choose *Web Service Configuration*.
3. Search for the corresponding consumer proxy, for example, `CO_FKKCC_SUBSCRIBER_ACCOUNT` (see figure below).

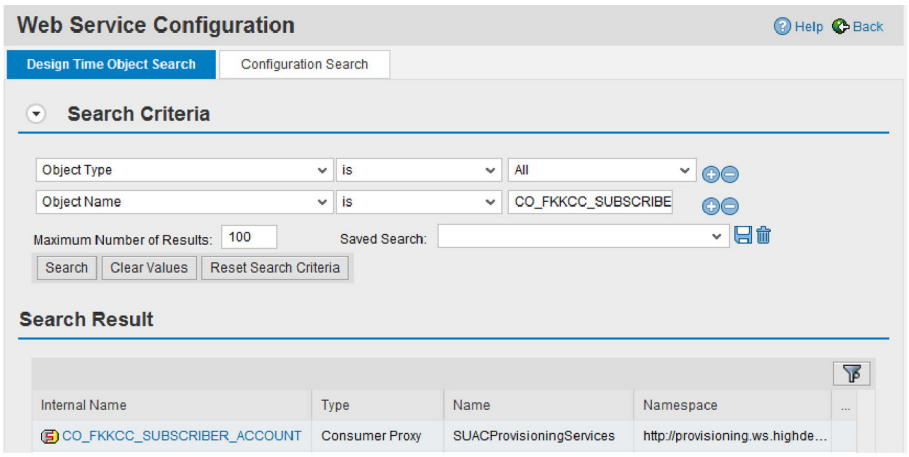


Figure 22: Web Service Configuration: SOA Manager

4. Select the entry.
5. Go to the *Configurations* tab.
6. Choose *Create > WSDL Based Configuration* to create a new logical port.
7. In the following wizard specify the corresponding logical port (see figure below):
   - Specify the logical port name.
   - Specify a description for the logical port.
   - Activate the *Logical Port is Default* checkbox.
   - Continue entering settings by choosing *Next*.

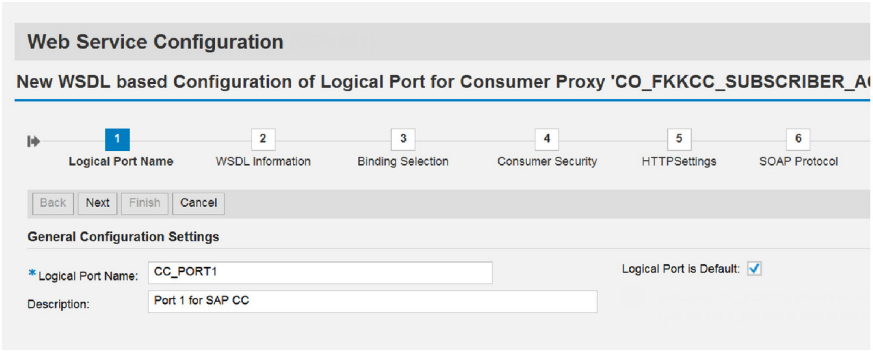


Figure 23: General Configuration Settings in Logical Port Wizard

8. Specify the WSDL information (see figure below).
   - As *WSDL Access Settings*, select the value *Via HTTP Access*.

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- Specify the URL for the WSDL, which is published on the corresponding SAP CC Core Server system (see table specifying the URL templates above).
- Continue entering settings by choosing *Next*.

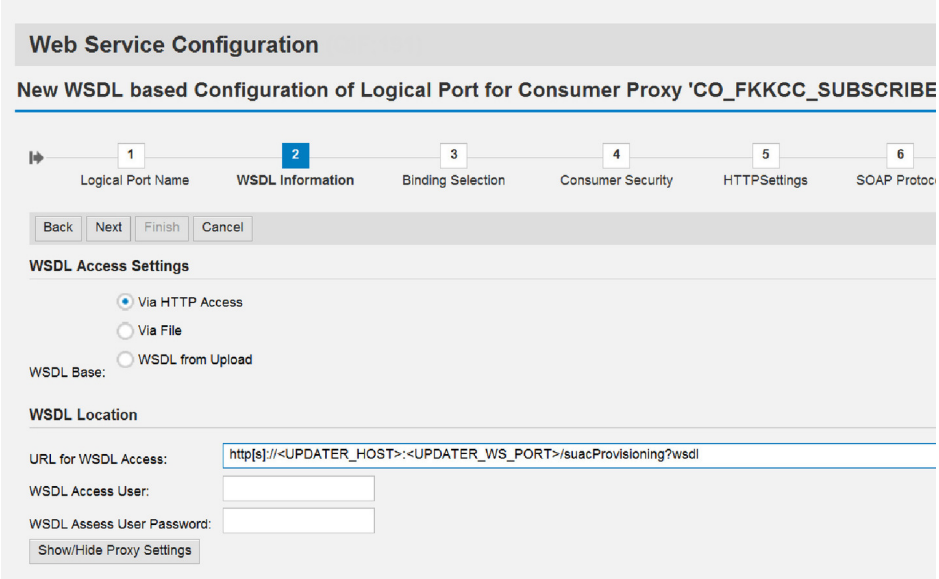


Figure 24: WSDL Information in Logical Port Wizard

9. Choose the port binding.

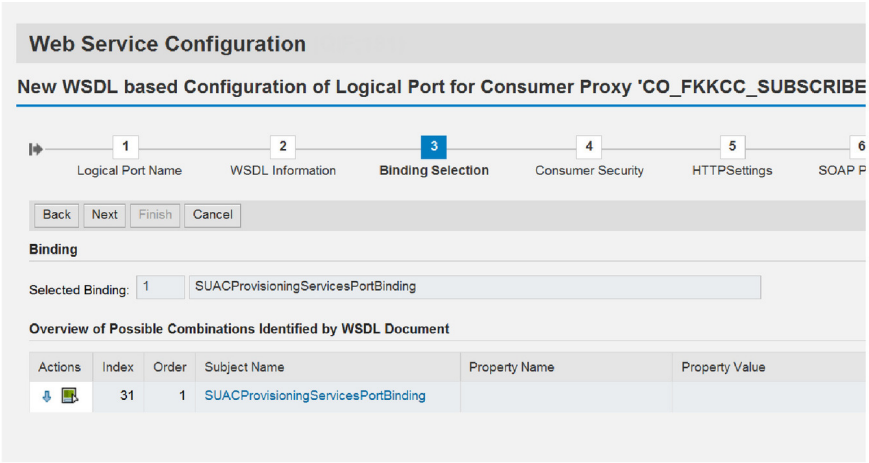


Figure 25: Binding Selection in Logical Port Wizard

- If only one binding is displayed in the list (see figure above), continue by choosing *Next*. If there is more than one listed, select the combination where the property name *Authentication Method is wsse:UsernameToken*.
- Continue entering settings by choosing *Next*.

10. Specify a user name and a password for the communication with the SAP CC system (see figure below).

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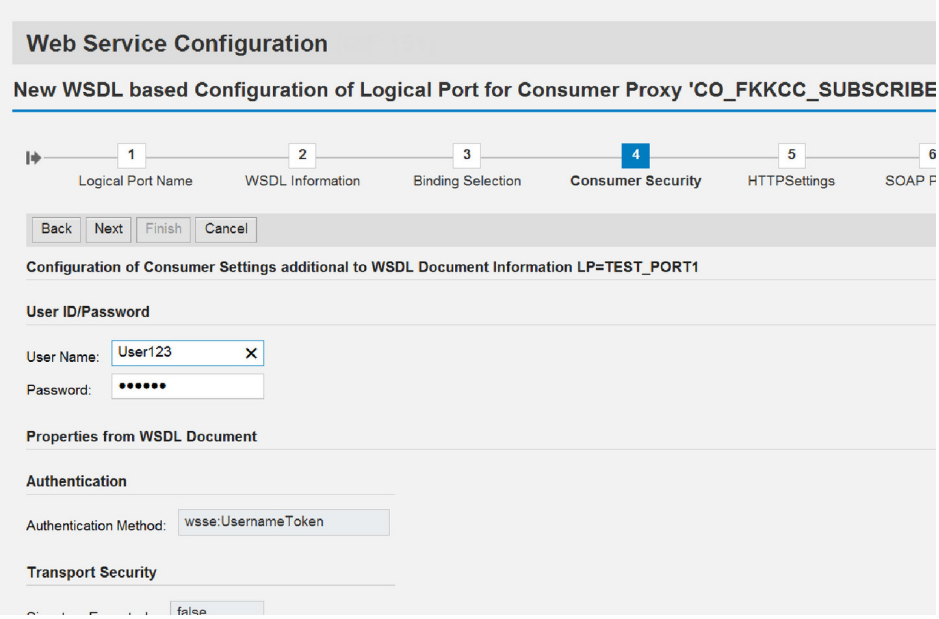


Figure 26: Consumer Security in Logical Port Wizard

11. Apply your settings by choosing *Next* and then *Finish*.

> **Note**
>
> **Integration into Single Sign-On Environments in SAP CC 4.1**
>
> In addition to the default Username Token security profile, the SAP Convergent Charging software also supports an Application to Application SSO mechanism to authenticate users consuming Web Services. This SSO mechanism only concerns the SOAP over HTTP communication channel, and is based on the SAML Token security profile which consists in transporting an SAML token into the header of SOAP messages.
>
> For more security information, see the Security Guide for SAP CC 4.1 on SAP Service Marketplace at service.sap.com/instguidescc > *Plan* > *Security Guide*.
>
> To configure logical ports with SAML token authentication, select the combination where the property name Authentication Method is *wsse: SAML Assertion*.

> **Note**
>
> For more information about user authentication and integration into SSO environments, see the SAP NetWeaver Security Guide at service.sap.com/securityguides > *SAP NetWevaer*.

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13 Setting Up Communication with SAP Convergent Charging Using RFC

To set up RFC communication between S/4 HANA and SAP CC systems, create a JCo user with the required authorizations in each S/4 HANA system.

To set up RFC communication between CRM and SAP CC systems, create a JCo user with the required authorizations in each CRM system.

To call an RFC function module in an S/4 HANA or CRM system from SAP Convergent Charging, set up the JCo communication in the SAP CC system. For more information, see section “System Settings for JCo Communication” in the SAP CC Configuration Guide under help.sap.com/cc > *SAP CC 4.1* > *Configuration and Deployment Information* > *SAP CC Configuration Guide* . Here you find information on how to configure the necessary JCo destinations and customer management areas.

> **Note**
>
> When you are connecting multiple FI-CA systems to one SAP CC system, in SAP CC you define a reference FI-CA system from which SAP CC uses all Customizing settings (settings for consumption item classes, billable item classes, and currencies). Make sure that this FI-CA reference system contains all Customizing settings applied to all the FI-CA systems you are connecting the SAP CC system to.
>
> **Example**
>
> You are using five FI-CA systems. Four systems are dedicated to the billing and invoicing of your customers. One system is dedicated to the billing and invoicing of the partners you are cooperating with (partner settlement). When you define the FI-CA reference system in SAP CC, this reference system has to contain the Customizing settings required for the billing and invoicing of your customers as well as your partners.

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14 Maintaining SAP CC Mapping and Range Tables Remotely

While configuring products or maintaining master agreements in SAP CRM, for example, you assign mapping and range tables from SAP CC used during rating and charging. From the CRM system you can search, read, change, and create the required mapping and range tables in SAP CC.

You can reference the identifier of a table in a charge plan using a parameter. The system uses the value of this parameter as the default value. However, when you create a provider contract in SAP CC, you can use a parameter value (table identifier) that differs from the default value.

In the CRM system you can overwrite charge plan parameters in the cross-catalog mapping of a CRM product. For parameters of the mapping or range table type, you can create a different table and assign its identifier to a parameter value. When you use a CRM product in a provider contract or master agreement, you can overwrite the table parameter defined in SAP CC. During the contract creation, the CRM system determines the appropriate parameter value (mapping or range table identifier) and sends it to the SAP CC system.

When you create content of mapping or range tables from the CRM system in SAP CC, you can also change the content from the CRM system. The content of the tables is not persisted in the CRM system. The CRM system only saves the table identifiers (table class ID and table ID) as default values or redefined parameter values.

SAP provides Web services for reading, creating, and maintaining SAP CC mapping and range tables. Configure the following consumer proxies of these Web services as described in the section “Setting Up Communication with SAP CC Using Web Services”:

- `CO_CRM_ISX_CC_CATALOG_SERVICES`
- `CO_CRM_ISX_CC_SUBSCRIBER_MAPPI`
- `CO_CRM_ISX_CC_SUBSCRIBER_RANGE`

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15 Setting Up Communication between SAP CRM and SAP S/4 HANA Using RFC

To establish RFC communication between the CRM and S/4 HANA system:

1. in Customizing for Customer Relationship Management under *CRM Middleware and Related Components* > *Communication Setup*:
   - Define RFC destinations
   - Create RFC users
   - Set up Logical Systems in the activities *Define Logical System* and *Assign Logical System to Client*
2. Using transaction `SM30` maintain the mapping between RFC destination and logical system in the view `CRMC_ABOXLSYSRFC`.

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16 Revenue Recognition

You can integrate the Offer-to-Cash end-to-end process with the SAP Revenue Accounting and Reporting add-on.

SAP Revenue Accounting and Reporting provides functions to support you with adhering to revenue recognition regulations, especially the regulations effective in countries subject to both US GAAP and IFRS.

SAP Revenue Accounting and Reporting automates the revenue recognition and accounting process and simplifies the tasks of revenue accountants in following the accounting guidelines. These tasks are described in the following steps:

- Identify the revenue contract (combine contracts)
- Identify the performance obligations in the contract
- Determine the transaction price
- Allocate the transaction price to the performance obligations of the contract
- Recognize the revenue as the performance obligations are satisfied

To enable the integration, you must activate the business function `FICA_EHP7_RA` in the S/4 HANA system and the business function `CRM_PROVORDERINT_3_9` in the CRM system.

Within the Offer-to-Cash end-to-end process, all data required by Revenue Accounting is communicated via the S/4 HANA system.

To integrate the Offer-to-Cash end-to-end process with Revenue Accounting, SAP provides the sender component `CA` (SAP O2C) with the SAP Revenue Accounting and Reporting add-on.

To establish communication between your S/4 HANA system and the Revenue Accounting system:

1. In the S/4 HANA system, activate the integration in Customizing for Contract Accounts Receivable and Payable under *Integration* > *Revenue Accounting*.

   In the same Customizing activity, also enter the RFC destination for the Revenue Accounting system.
2. In the Revenue Accounting system, define logical systems and assign these to the sender component `CA` (SAP O2C) (see Customizing for Revenue Accounting under *Inbound Processing* > *Revenue Accounting Item Management*).

Revenue Accounting receives the operative data (order items, invoice items and fulfillment items) from the S/4 HANA system and stores the data in the form of revenue accounting items. The technical properties of the revenue accounting items are determined by revenue accounting item classes. You define these technical properties in the Customizing for Revenue Accounting under *Inbound Processing* > *Revenue Accounting Items*.

In order to transfer revenue accounting items from the S/4 HANA system to the revenue accounting system, the revenue accounting item classes `CA01` (for order items), `CA02` (for fulfillment items), and `CA03` (for invoice items) must be available in the revenue accounting system. SAP defines the technical names of these classes. Using these technical names during revenue item class generation, the system automatically applies the settings required for each class.

For more information, see the following sources:

- SAP Revenue Accounting and Reporting documentation at help.sap.com/revacc
- Documentation for business functions `CRM_PROVORDERINT_3_9` and `FICA_EHP7_RA`
- SAP CRM documentation at help.sap.com/crm704 > *Application Help* > *English* > *SAP CRM for Industries* > *Telecommunications* > *Functions for Provider Sales and Order Management* > *Provider Order*

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and Provider Contract > *Provider Contracts* > *Replication of Provider Contracts to SAP S/4 HANA and SAP CC* >

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17 Authorizations

SAP CRM, SAP S/4 HANA, and SAP CC use different authorization concepts. While SAP CRM and SAP S/4 HANA grant authorizations per business object or business function using authorization objects, SAP CC uses a more general concept based on roles. No one-to-one relationship between the different components exists for authorizations. Therefore, SAP does not provide a mapping for authorizations across systems within the integrated business process.

The following table lists all S/4 HANA and CRM authorization objects and, where possible, provides you with options on how to map between the two within your project.

Table 16

Category	Object or Function Protected	Authorization Object in SAP S/4 HANA	Authorization Objects in SAP CRM	Options for Cross-System Authorization Checks
Master data object	• Contract account • Business agreement	• F_KKVK_BEG (authorization group) • F_KKVK_BUK (company code) • F_KKVK_VKT (contract account type) • F_KKVK_FDG (field groups in the Business Data Toolset (BDT))	CRM_BA_CLS	
Master data object	Prepaid account	• F_KKVK_BEG (using function module FKK_ACCOUNT_AUTHORITY_CHECK) • F_KKVK_BUK (using function module FKK_ACCOUNT_AUTHORITY_CHECK) • F_KKPA_AUT (BDT authorization types) • F_KKPA_FDG (BDT field groups)	Subobject of the business agreement	
Master data object	• Provider contract • Partner agreement	• F_KKVT_BEG (authorization group) F_KKVT_BUK (company code) • F_KKVT_AUT (BDT authorization types)	• CRM_ORD_LP (visibility in organizational model) • CRM_ORD_TE (visibility in territory;	The S/4 HANA system derives the company code from the sales organization in SAP CRM. So you can align authorization checks for the company code

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		• F_KKVT_FDG (field groups in the BDT)	must be activated for transaction type) • CRM_ORD_PR (business transaction type) • CRM_ORD_OE (allowed organizational units) • BAdI CRM_ORDER_AUTH_CHECK (additional checks or alternative checks instead of standard checks) • customer checks using the Access Control Engine (ACE)	and sales organization across the two systems using the authorization objects F_KKVT_BUK in the S/4 HANA system and CRM_ORD_OE in the CRM system. If you consider the transaction type when determining the sales organization in the CRM system, you can implicitly link the authorizations for the company code in the S/4 HANA system and the authorizations for the transaction type in the CRM system. As the BEGRU field is not available in the CRM system, you can, for example, use the sales group field (part of CRM_ORD_OE) to implement authorization checks. The sales group can act as a substitute for the BEGRU field, which you can replicate (custom solution). As an alternative, you can substitute a custom field for the BEGRU field in the CRM system, and implement an authorization check for this custom field using the BAdI CRM_ORDER_AUTH_CHECK. Make sure, that the determined field value is transferred to the BEGRU field in the S/4 HANA system.

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Master data object	Provider order	none	• CRM_ORD_LP (visibility in organizational model) • CRM_ORD_TE (visibility in territory; must be activated for transaction type) • CRM_ORD_PR (business transaction type) • CRM_ORD_OE (allowed organizational units) • BAdI CRM_ORDER_AUTH_CHECK (additional checks or instead of standard checks) • Customer checks using the Access Control Engine (ACE)	
Master data	Provider contract message	X_COL_OBJ (Cross-application authorization objects)	not applicable	
Master data object	Product	Authorization based on the transaction code	• COM_PRD (general product authority) • COM_PRD_CT (product category specific)	
Master data object	Product (during product transport from SAP CRM to SAP CRM)	not applicable	In the source system the user requires the authorization for authorization object CRM_IU_PDL (Utilities Product download). In the target system the user requires the authorization for the following authorization object: • CRM_IU_PDL (Utilities Product download) • S_RFC for function group	

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			CRM_IU_PRODUCT_TRANSPORT	
Master data object	Master agreement	• F_KKMA_BEG (authorization group) • F_KKMA_BUK (company code) • F_KKMA_AUT (authorization types in the BDT) • F_KKMA_FDG (field groups in the BDT)	• CRM_ORD_LP (visibility in organizational model) • CRM_ORD_TE (visibility in territory; must be activated for transaction type) • CRM_PRVMA (business object provider master agreement) • CRM_ORD_PR (business transaction type) • CRM_ORD_OE (allowed organizational units) • BAdI CRM_ORDER_AUTH_CHECK (additional checks or alternative checks instead of standard checks) • Customer checks using the Access Control Engine (ACE)	If you consider the transaction type when determining the sales organization in the CRM system, you can implicitly link the authorizations for the company code in the S/4 HANA system and the authorizations for the transaction type in the CRM system. As the BEGRU field is not available in the CRM system, you can, for example, use the sales group field (part of CRM_ORD_OE) to implement authorization checks. The sales group can act as a substitute for the BEGRU field, which you can replicate (custom solution). As an alternative, you can substitute a custom field for the BEGRU field in the CRM system, and implement an authorization check for this custom field using the BAdI CRM_ORDER_AUTH_CHECK. Make sure, that the determined field value is transferred to the BEGRU field in the S/4 HANA system. If you cannot set the BEGRU field in the CRM system, you have to supplement the newly created master agreements with the

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				BEGRU value in the S/4 HANA system. The system performs the authorization check only when the BEGRU field is filled. You achieve this, by setting the revision status *3* (Newly created, billing and invoicing blocked) using the function module FKK_MA_SAMPLE_0530 for FI-CA event 0530. You can then select the newly created master agreements with status *3* in transaction FP_MA0 and supplement the value of the BEGRU field manually. Instead of setting a value for the BEGRU field manually, you can also use FI-CA event 0530 to directly set the value during replication.
Transaction	FP40P (*Process Prepaid Balance*)	• F_PREP_BEG (authorization group) • F_PREP_BUK (company code) You can customize authorization levels for prepaid actions. The system applies them, as well.	not applicable (in Financial Customer Care)	
Master data object	Master data replication to SAP CC	Authorization based on the transaction code	not applicable	
Transaction	Financial Customer Care (FCC)	Authorization check for transaction code FKKCRM_AUTH_CHECK (using function module FKK_CRM_GENERAL_AUTH_CHECK)	Role based authorization checks as is the standard in SAP CRM	

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		Backend authorization checks are carried out for the respective functionality.		
Customizing	Configuration of billable item classes and consumption item classes	F_KKBIXCON	not applicable	
Customizing	Generation of billable item classes and consumption item classes	F_KKBIXGEN	not applicable	
Process	Billing and billing documents	F_KKINVBIL (considers BEGRU of the contract account)	not applicable	
Process	Invoicing and invoicing documents	F_KKINV (considers VKTYP and BEGRU of the contract account)	Display of invoices in Financial Customer Care uses backend authorization checks in S/4 HANA	
Transactional data	Billable item	• F_KKBIXBIT (only status BIT0) • F_KKCIBIT2 (for creating, changing, displaying) • F_KKBIXDEL (for deleting) • F_KKBIXEXC (exception)	When creating billable items as one-offs, S/4 HANA backend authorizations are checked	
Transactional data	Chargeable item	F_KKBIXCIT (for creating, changing, displaying)	not applicable	
Transactional data	Display of billable Items in monitor	F_KKBIXMON	Display of billable items in Financial Customer Care uses backend authorization check of S/4 HANA	
Transactional data	Correction Processes	• F_KKBIXEXC (exempt) • F_KKBIXEXC (restore) • F_KKBIXBIT and F_KKCIBIT2 (create) • F_KKBIXMON and F_KKCIBIT2 (display)	not applicable	

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Transactional data	Creation of test data	F_KKBIXTST	not applicable	
Transactional data	Dispute (one order object)	Authorization objects of case management (as for example, authorization object S_SCMG_CAS) are checked when maintaining dispute cases	• CRM_ORD_LP (visibility in organizational model) • CRM_ORD_TE (visibility in territory; must be activated for transaction type) • CRM_CMP (business object: complaints) • CRM_ORD_PR (business transaction type) • CRM_ORD_OE (allowed organizational units) • BAdI CRM_ORDER_AUTH_CHECK (additional checks or alternative checks instead of standard checks) • Customer checks using the Access Control Engine (ACE)	

For Information about authorizations within the integrated business process using SAP S/4 HANA, SAP CRM, and SAP CC, see the following sections of the security guide for SAP CC on SAP Help Portal at help.sap.com/cc41 ▶ *Security Information*:

- User Administration and Authorization
- Authorizations

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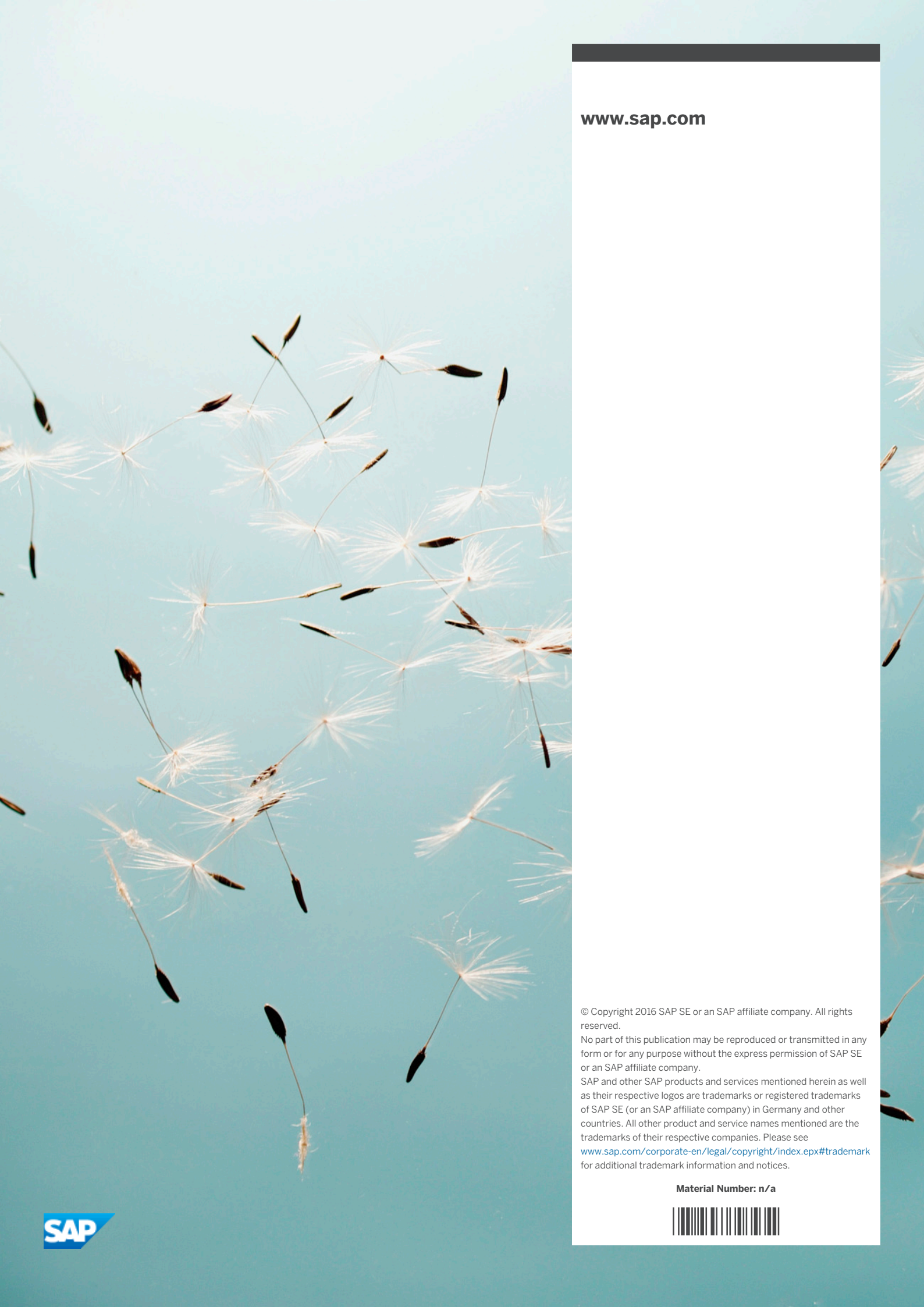
Typographic Conventions

Table 17

Example	Description
<Example>	Angle brackets indicate that you replace these words or characters with appropriate entries to make entries in the system, for example, “Enter your **<User Name>**”.
Example > *Example*	Arrows separating the parts of a navigation path, for example, menu options
Example	Emphasized words or expressions
Example	Words or characters that you enter in the system exactly as they appear in the documentation
www.sap.com	Textual cross-references to an internet address
/example	Quicklinks added to the internet address of a homepage to enable quick access to specific content on the Web
123456	Hyperlink to an SAP Note, for example, SAP Note 123456
Example	• Words or characters quoted from the screen. These include field labels, screen titles, pushbutton labels, menu names, and menu options. • Cross-references to other documentation or published works
`Example`	• Output on the screen following a user action, for example, messages • Source code or syntax quoted directly from a program • File and directory names and their paths, names of variables and parameters, and names of installation, upgrade, and database tools
`EXAMPLE`	Technical names of system objects. These include report names, program names, transaction codes, database table names, and key concepts of a programming language when they are surrounded by body text, for example, `SELECT` and `INCLUDE`
EXAMPLE	Keys on the keyboard

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